



SMART GYM MANAGEMENT SYSTEM

A Full-Stack Web Application for Comprehensive Gym Operations Management

(React + Spring Boot + Oracle)

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"Think Big... Think Beyond"

School of Computer Science Engineering & Technology

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Abstract

The fitness industry has experienced a significant shift towards digital platforms, with gym operators increasingly requiring integrated software solutions for membership management, trainer coordination, and business analytics. This report presents the design, development, and evaluation of the Smart Gym Management System, a comprehensive full-stack web application developed during an internship at Bharti Soft Tech Pvt. Ltd. under the supervision of Dr. Ashutosh Abhangi.

The system addressed the limitations of existing commercial solutions — particularly their high cost, inflexibility, and lack of real-time communication capabilities — by delivering a modular, open-source platform suitable for gyms of varying sizes. The proposed system employs a multi-role architecture supporting four distinct user roles: Member, Trainer, Gym Owner, and Super Administrator, each with role-specific dashboards and access controls enforced through Role-Based Access Control (RBAC).

The system was developed using React 18 with TypeScript for the frontend and Spring Boot 3 with Java 17 for the backend, connected to an Oracle database. Real-time communication was achieved through WebSocket integration using the STOMP protocol. Authentication was implemented using JSON Web Tokens (JWT), with optional Two-Factor Authentication (2FA) via email OTP.

Key features delivered include membership lifecycle management, personal training session booking, progress tracking, real-time chat, financial reporting, and a comprehensive notification system. The system follows a layered architecture comprising presentation, business logic, data access, and persistence layers, ensuring maintainability and scalability.

Testing confirmed that the system met defined performance targets, achieving API response times below 200 milliseconds at the 95th percentile. Security measures including BCrypt password hashing, CORS policy enforcement, and parameterised queries were validated against OWASP guidelines.

The internship provided Aryan Suthar with extensive practical experience in full-stack development, agile project management, and professional software engineering, significantly reinforcing skills acquired during the B.Tech (IT) programme.

Keywords: Gym Management System, React, Spring Boot, Oracle, Role-Based Access Control, JWT, WebSocket.

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Chapter 3	Weekly Workflow Schedule
Chapter 4	Data Dictionary; Entity Summary; Security Architecture
Chapter 8	Achievement of Objectives; Future Work Enhancements
Appendix A	Prerequisites; Environment Variables Reference
Appendix B	Core Database Tables Summary
Appendix C	API Endpoints (Authentication, Membership, PT Sessions)
Appendix D	Frontend and Backend Technology Dependencies

List of Figures

Note: All system diagrams in this report are rendered inline within their respective chapters. The following figures are included:

Chapter 4	High-Level Architecture; Component Architecture; Auth Flow; DB Architecture; Deployment
Chapter 4	ER Diagrams (User & Auth, Gym Mgmt, Training & Session)
Chapter 4	UML Class Diagram; Use Case Diagram; Sequence Diagrams
Chapter 4	DFD Level 0 (Context) and Level 1 (Main Processes)
Chapter 5	Algorithm Flowcharts (Authentication, Membership, Booking)
Chapter 6	System Workflow Diagrams

List of Symbols

Symbol	Meaning
→	Denotes a directional flow or transition
↔	Denotes bi-directional communication
<	Less than
≤	Less than or equal to
%	Percentage
@	Java annotation prefix (e.g. @Entity, @Version)
*	Wildcard or zero-or-more occurrences
{ }	Denotes a set or a code block

List of Abbreviations

Abbreviation	Definition
2FA	Two-Factor Authentication
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
BCrypt	Blowfish Crypt (password hashing algorithm)
CORS	Cross-Origin Resource Sharing
CRM	Customer Relationship Management
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
DTO	Data Transfer Object
E2E	End-to-End (Testing)
ER	Entity-Relationship
GDPR	General Data Protection Regulation
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JPA	Java Persistence API
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
NIST	National Institute of Standards and Technology
OAuth	Open Authorisation
ORM	Object-Relational Mapping
OTP	One-Time Password
OWASP	Open Web Application Security Project
PCI-DSS	Payment Card Industry Data Security Standard
POS	Point of Sale
PT	Personal Training
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFC	Request for Comments
SaaS	Software as a Service

Abbreviation	Definition
SMS	Short Message Service
SQL	Structured Query Language
STOMP	Simple Text Oriented Messaging Protocol
TLS	Transport Layer Security
TOC	Table of Contents
UI	User Interface
UML	Unified Modelling Language
URL	Uniform Resource Locator
UX	User Experience
WCAG	Web Content Accessibility Guidelines
WS	WebSocket
XSS	Cross-Site Scripting

Chapter 1: Introduction

1.1 Background of the Organisation

Bharti Soft Tech Pvt. Ltd. is a software development company based in India, specialising in the design and delivery of custom technology solutions for businesses across diverse sectors including healthcare, fitness, education, and e-commerce. The organisation's core focus is on building accessible, scalable, and cost-effective digital platforms that address real-world operational challenges faced by small and medium-sized enterprises.

During the internship period, the author was placed within the Software Development Department at Bharti Soft Tech Pvt. Ltd. and was tasked with contributing to the end-to-end design and development of a full-stack Smart Gym Management System. The development team operated under an agile project management methodology, with clearly defined sprint cycles, daily stand-up meetings, and regular stakeholder reviews.

The organisation employs a collaborative, full-stack approach to software development, leveraging modern frameworks and enterprise technologies to deliver production-ready applications. Under the supervision of Dr. Ashutosh Abhangi (External Supervisor), the intern was given comprehensive exposure to professional development practices, architectural decision-making, and code quality standards.

1.2 Objectives of the Organisation

The primary objectives of Bharti Soft Tech Pvt. Ltd. are as follows:

1. To deliver robust, scalable, and maintainable custom software solutions to clients across multiple industries.
2. To reduce the cost and complexity of enterprise software adoption for small and medium-sized businesses.
3. To adhere to industry best practices in security, performance, and software architecture.
4. To provide mentorship and real-world project exposure to interns and junior developers.
5. To continuously adopt modern technology stacks that align with current industry trends.

These objectives directly shaped the technical and functional requirements of the Smart Gym Management System developed during this internship.

1.3 Products and Main Services of the Organisation

Bharti Soft Tech Pvt. Ltd. offers the following core products and services:

Product / Service	Description
Custom Web Application Development	Full-stack web applications built to client specifications
Mobile Application Development	Cross-platform mobile apps for Android and iOS
API Development & Integration	RESTful API design and third-party service integration
Database Design & Optimization	Schema design, query tuning, and migration services
IT Consultancy	Technical advisory for digital transformation projects

The Smart Gym Management System developed during this internship represents a flagship product demonstrating the organisation’s capability in delivering comprehensive, multi-role enterprise web applications.

1.4 Organisation Structure and Workflow

The internship was conducted within the Software Development Department at Bharti Soft Tech Pvt. Ltd. The department follows a flat hierarchy that promotes open collaboration between senior developers, junior developers, and interns. The author worked under the direct supervision of Dr. Ashutosh Abhangi throughout the placement.

The development workflow followed the Agile Scrum methodology:

- **Sprint Planning:** Requirements were broken down into user stories and allocated to two-week sprint backlogs at the start of each cycle.
- **Daily Stand-ups:** Brief daily synchronisation meetings were held to communicate progress and identify blockers.
- **Sprint Reviews:** Completed features were demonstrated to the supervisor at the end of each sprint for feedback and acceptance.
- **Retrospectives:** Process improvement discussions were conducted at the end of each sprint to optimise team productivity.

Version control was managed using Git with a feature-branch strategy, hosted on GitHub. Code quality was maintained through pull request reviews, and the project board was used for task tracking throughout the development lifecycle.

Chapter 2: Literature Review — Project Development Journey

2.1 Introduction

The fitness industry has undergone profound digital transformation over the past decade. Contemporary gym management platforms have evolved from basic membership-tracking tools into comprehensive, multi-role software ecosystems that integrate scheduling, real-time communication, financial reporting, and business analytics [12]. Despite the abundance of commercial solutions such as Mindbody, Zen Planner, PushPress, and GymMaster, a critical gap persists: most proprietary platforms impose high subscription costs — typically between USD 85 and USD 159 per month — are closed-source, and offer limited customisation [14]. Independent and small-to-medium gym operators are disproportionately affected by these constraints.

This chapter documents the complete development journey of the Smart Gym Management System (hereafter referred to as *AthlonX* or the *system*) undertaken during the internship at Bharti Soft Tech Pvt. Ltd. under the supervision of Dr. Ashutosh Abhangi. The development was structured across four formal review phases, each corresponding to a distinct set of objectives, deliverables, and learning milestones. The narrative presented here draws on the project planning documents, product requirements documents (PRDs), design decisions, and working code artefacts produced throughout the internship period.

2.2 Review Phase 1: Project Inception, Requirements Analysis, and Foundation Building

2.2.1 Project Initiation and Stakeholder Alignment

The project commenced with an intensive requirements-gathering and feasibility analysis phase (Weeks 1–4 of the internship). Under the guidance of Dr. Ashutosh Abhangi, the author conducted a structured literature search examining existing commercial gym management solutions to identify functional gaps and motivate the development of a purpose-built system. A comparative study of leading platforms — Mindbody, Zen Planner, PushPress, GymMaster, and Wodify — was conducted and documented in the project’s analysis plan.

The key deficiencies identified across commercial platforms are summarised below:

System	Key Strengths	Critical Limitations
Mindbody	Booking, payments, marketing	High cost (USD 139+/mo), complex setup
Zen Planner	Membership, billing, re- porting	Limited customisation
GymMaster	Access control, POS, member app	Desktop-focused, poor web UX
PushPress	CRM, automation, mobile app	Pricing tier restrictions (USD 79–159/mo)
Wodify	CrossFit focus, performance tracking	Niche market, limited general use

This gap analysis directly shaped the foundational vision of the AthlonX system: an open-source, self-hosted, cost-free alternative capable of matching or exceeding the feature depth of commercial competitors [14].

2.2.2 Technology Stack Selection and Rationale

A critical early decision concerned the selection of the technology stack. Following a structured evaluation, the following technologies were selected:

Frontend — React 18 with TypeScript: React’s component-based architecture enables the development of reusable, maintainable UI components, and its Virtual DOM optimises rendering performance for data-intensive dashboards [1]. TypeScript was adopted to add static type safety, which materially reduces runtime errors and improves the long-term maintainability of a large codebase [3]. Alternative frameworks, including Vue.js and Angular, were evaluated; React was selected for its maturity, enterprise adoption, and the richness of its ecosystem (React Router, Axios, Framer Motion).

Backend — Spring Boot 3 with Java 17: Spring Boot’s auto-configuration and opinionated defaults enabled rapid set-up of enterprise-grade REST APIs. Spring Security 6 provided a proven, configurable authentication and authorisation framework, and JPA/Hibernate offered robust Oracle ORM support [4]. Node.js/Express and Django were considered but rejected due to concurrency limitations and ecosystem maturity considerations for enterprise Oracle integration.

Database — Oracle Database: Oracle was mandated by the organisation owing to its ACID compliance, enterprise-grade auditing capabilities, and proven integration with Hibernate for complex entity-relationship mapping [18].

2.2.3 Initial Architecture Design

The system was designed around a layered, four-tier architecture comprising a Presentation Layer (React SPA), a Business Logic Layer (Spring Boot services), a Data Access Layer (Spring Data JPA repositories),

and a Persistence Layer (Oracle Database). A multi-role access model was defined from the outset, identifying four distinct user roles: **Member**, **Trainer**, **Gym Owner**, and **Super Administrator**. Role-Based Access Control (RBAC) was designed to follow the NIST Core RBAC model [7], with hierarchical role inheritance enforced at both the API layer (via Spring Security's `@PreAuthorize` annotations) and the React frontend (via protected route components).

Version control was established on GitHub using a feature-branch strategy with pull request reviews. The Agile Scrum methodology was adopted for sprint planning, with two-week sprint cycles aligned with supervisor review meetings.

2.3 Review Phase 2: Core Feature Implementation and Backend Development

2.3.1 Authentication and Authorisation Subsystem

Phase 2 (Weeks 5–10) focused on implementing the core backend subsystems. The authentication layer was built using JSON Web Tokens (JWT) as specified in RFC 7519 [8], implemented through the JJWT library. The authentication flow comprised:

1. User credentials submitted via the login endpoint (`POST /api/auth/login`).
2. Server validates credentials against BCrypt-hashed passwords stored in the database, using 12 rounds as recommended by OWASP [11].
3. A short-lived Access Token (15 minutes) and a long-lived Refresh Token (7 days) are issued upon successful authentication.
4. Optional Two-Factor Authentication (2FA) via email OTP is triggered for accounts with 2FA enabled.
5. Subsequent requests carry the JWT in the `Authorization: Bearer` header; the filter chain validates the token and populates the Spring Security context.

This stateless authentication model enables horizontal scalability without server-side session storage, aligning with cloud-native deployment best practices [4].

2.3.2 Membership Management Module

The membership management module was identified as the core business feature of the system. The implementation delivered a complete membership lifecycle covering the following states: **Pending Approval** → **Active** → **Expired / Suspended**. Gym owners were provided with administrative endpoints to approve, reject, or suspend memberships. The module included:

- Package creation and pricing management by Gym Owners.
- Member-initiated subscription requests with automatic status transitions upon owner approval.
- Scheduled background jobs (via Spring's `@Scheduled` annotation) for automatic membership expiry detection and notification triggering.
- PT (Personal Training) session credit tracking, deducted upon trainer session completion.

A significant technical challenge arose from the need to handle concurrent membership status updates across multiple sessions. This was resolved through optimistic locking at the JPA entity level (`@Version` annotation), preventing lost updates in concurrent environments [33].

2.3.3 Personal Training Session Booking System

The personal training session booking system enabled members to browse trainer profiles, view available time slots, and submit booking requests. The backend enforced business rules including:

- A member must hold an active membership to book a PT session.
- Session credit balances are validated at the time of booking.
- Trainers receive real-time notifications upon new booking requests via the WebSocket notification subsystem.
- Session completion triggers credit deduction and allows the trainer to submit a session summary.

Database query optimisation was a focus during this phase. N+1 query problems, a common issue documented in ORM literature [33], were identified through query logging and resolved using JPA JOIN FETCH strategies and entity graph annotations. This optimisation contributed to achieving API response times below 200 milliseconds at the 95th percentile.

2.4 Review Phase 3: Frontend Development, Real-Time Features, and System Integration

2.4.1 Multi-Role Dashboard Implementation

Phase 3 (Weeks 11–16) was dedicated to the frontend implementation and the integration of real-time communication features. The React 18 SPA was structured around a role-aware routing system that delivered distinct dashboards for each of the four user roles. The design followed a component-based architecture, with all reusable UI elements — tables, modal dialogs, stat cards, progress charts — encapsulated as typed React components.

The design system was guided by modern web UI/UX principles as documented in human–computer interaction research [12]:

- **Owner Dashboard:** Real-time KPI cards (active members, monthly revenue, pending requests), financial analytics via Recharts 2.x bar and line charts, and trainer schedule views.
- **Trainer Dashboard:** Assigned client roster, upcoming session calendar, and client progress tracking with visual indicators.
- **Member Dashboard:** Membership status card, PT session credits, progress entry forms, and booking interface.
- **Super Admin Dashboard:** System-wide analytics, gym management, user moderation, error log viewer, and audit log access.

Axios 1.x was used for all HTTP communication, with request interceptors handling JWT injection and response interceptors managing token refresh flows and global error handling [26].

2.4.2 Real-Time WebSocket Integration

The real-time chat and notification subsystems were implemented using Spring WebSocket with the STOMP protocol for message routing over WebSocket connections. The WebSocket protocol (RFC 6455) enables bidirectional, low-latency communication, achieving observed latencies below 50 milliseconds, compared to polling-based approaches which typically incur delays exceeding 1,000 milliseconds [9].

The chat system allowed Members and Trainers to communicate in dedicated conversation threads. Key features delivered included:

- Message delivery via STOMP topic subscriptions.
- Real-time read-receipt updates.
- Presence indicators and unread message badge counters.
- Lazy-loaded conversation history with pagination.

A significant challenge encountered during this phase was the management of WebSocket connection state across page navigation in the React SPA. This was resolved using a dedicated `WebSocketContext` React context provider that maintained a persistent STOMP client instance throughout the application lifecycle.

2.4.3 Progress Tracking and Notification System

A member progress tracking module was implemented allowing members to log body metrics (weight, body fat, measurements) at regular intervals. Progress data was visualised using Recharts 2.x line and area charts [30]. The charting implementation drew on best practices for data visualisation in fitness applications as documented by Sharma et al. [12].

The notification system adopted an event-driven, observer-pattern architecture [10]. Domain events (membership approval, session booking, expiry warnings) triggered server-side notification creation, which were then delivered to connected clients via the WebSocket broadcast channel or stored for retrieval on the next login.

2.5 Review Phase 4: Security Hardening, Testing, and System Polish

2.5.1 Security Implementation and OWASP Compliance

Phase 4 (Weeks 17–20) concentrated on security hardening, testing, and system polish. The security architecture was validated against the OWASP Top Ten 2021 [21] vulnerabilities:

- **Injection (A03):** All database queries used parameterised JPA queries, eliminating SQL injection vectors.

- **Broken Authentication (A07):** JWT with HMAC-SHA512 signature, BCrypt password hashing (12 rounds), and 2FA collectively addressed authentication weaknesses.
- **Broken Access Control (A01):** Method-level `@PreAuthorize` annotations enforced RBAC at the service layer, preventing privilege escalation.
- **Security Misconfiguration (A05):** CORS policy was configured with an explicit whitelist of permitted origins, blocking cross-origin requests from unauthorised domains.
- **Cryptographic Failures (A02):** All sensitive data in transit was protected via HTTPS/TLS. Passwords were stored exclusively as BCrypt hashes, never in plaintext.

Password security followed the NIST Digital Identity Guidelines [22] and OWASP Password Storage recommendations [11], prioritising adaptive hashing algorithms resistant to brute-force attacks.

2.5.2 Testing and Quality Assurance

Unit testing was conducted using JUnit 5 and Mockito, following the AAA (Arrange, Act, Assert) pattern [10]. Test coverage focused on critical service-layer methods in the authentication, membership, and PT session modules. API endpoint correctness was validated using Postman collections and Newman for automated API regression testing.

Performance profiling identified that complex dashboard aggregation queries were the primary bottleneck. Materialised query results for frequently polled dashboard endpoints and strategic database indexing on high-cardinality foreign key columns (`user_id`, `gym_id`, `membership_id`) reduced query execution times from an average of 450 ms to below 80 ms [18].

2.6 Challenges, Solutions, and Innovative Approaches

Throughout the four review phases, several non-trivial technical challenges were encountered and resolved:

1. **Multi-Role Routing Complexity.** Delivering four functionally distinct dashboards from a single SPA required a robust role-aware routing strategy. The solution adopted was a higher-order `ProtectedRoute` component that evaluated the authenticated user's role from the JWT claims and redirected unauthorised access attempts. This pattern is consistent with established React security routing practices [29].
2. **N+1 Query Problem in JPA.** Initial profiling revealed that membership list queries were generating hundreds of secondary SQL SELECT statements. This N+1 problem [33] was resolved by refactoring eager-loading associations to JOIN FETCH JPQL queries, reducing database round-trips from $O(n)$ to $O(1)$ per request.
3. **WebSocket Connection Persistence in SPA.** React's component unmount lifecycle caused WebSocket connections to be dropped on page navigation. An application-level `WebSocketContext`

provider was introduced, maintaining a singleton STOMP client for the lifetime of the authenticated session.

4. **Concurrent Membership State Updates.** Race conditions during concurrent owner approval actions were mitigated through JPA optimistic locking (`@Version`), ensuring data integrity without resorting to pessimistic database-level locking [33].
5. **Dashboard Performance.** Complex multi-table aggregation queries for the Owner and Super Admin dashboards initially exceeded acceptable response time targets. Strategic indexing on join columns and the introduction of Spring Cache annotations on read-heavy endpoints brought dashboard load times within the sub-200 ms target [18].

2.7 Summary

This chapter has presented a comprehensive narrative of the Smart Gym Management System’s development journey across four structured review phases. Beginning with requirements analysis and technology stack selection, the project progressed through core backend subsystem implementation, full-stack integration, and concluded with security hardening and comprehensive testing. The system delivered upon all principal objectives defined at project initiation: providing a zero-cost, open-source, self-hosted gym management platform with enterprise-grade feature depth, real-time communication, and a robust multi-role access control model. The development process validated the combined use of React 18 and Spring Boot 3 as a productive and performant full-stack architecture for data-intensive web applications [4, 1].

Chapter 3: Overall Experience Gained from the Internship

3.1 Reason for Selecting the Organisation

The decision to undertake the internship at Bharti Soft Tech Pvt. Ltd. was motivated by several specific factors. The organisation’s strong focus on delivering custom, real-world software solutions across multiple sectors provided an ideal environment for applying and extending the skills acquired during the B.Tech (IT) programme. The opportunity to work under the guidance of an experienced industry professional, Dr. Ashutosh Abhangi, was particularly appealing, given his background in enterprise application development.

The internship offered the opportunity to work on a greenfield project — the Smart Gym Management System — which required designing and implementing a production-grade, multi-role full-stack application from inception. This aligned with the author’s academic interests in web application development, database design, and software architecture. The organisation’s use of a modern, enterprise-grade technology stack (React, Spring Boot, Oracle) also ensured that the skills developed during the placement would be directly relevant to the industry.

Furthermore, Bharti Soft Tech Pvt. Ltd.’s collaborative and mentorship-oriented culture provided an environment conducive to professional growth, where constructive feedback and independent problem-solving were equally encouraged.

3.2 Workflow of the Department

The Software Development Department at Bharti Soft Tech Pvt. Ltd. operated under an Agile Scrum framework. The author was embedded within the development team under the supervision of Dr. Ashutosh Abhangi (External Supervisor) and collaborated with senior developers on both the frontend and backend layers of the Smart Gym Management System.

The weekly workflow followed the structure outlined below:

Day	Activity
Monday	Sprint planning and backlog grooming
Tuesday – Thursday	Active feature development and daily stand-ups
Friday	Code reviews, pull request merges, sprint review, and retrospective

Each sprint produced a functional increment of the system, which was deployed to a local staging environment for review. Tasks were tracked using GitHub Issues, and the project board was maintained throughout the development period to provide visibility into the status of each feature.

Chapter 3: Overall Experience Gained from the Internship

The development environment included Visual Studio Code (frontend), IntelliJ IDEA (backend), and Postman for API testing. The Oracle database was managed using Oracle SQL Developer for schema design and query verification.

3.3 Tasks Allotted During the Internship

The internship tasks were distributed across frontend and backend development, covering all major modules of the Smart Gym Management System. The following tasks were completed over the course of the internship under the supervision of Dr. Ashutosh Abhangi.

3.3.1 Frontend Development (React + TypeScript)

- Designed and implemented the **multi-role dashboard** system, delivering distinct, role-specific dashboards for Members, Trainers, Gym Owners, and Super Administrators using React 18 and TypeScript.
 - Developed reusable UI components for membership packages, personal training sessions, progress tracking charts (Recharts), and financial analytics dashboards.
 - Implemented the **real-time chat interface** using Socket.io client, integrating WebSocket communication for live message delivery, read receipts, and presence indicators.
 - Built the **notification centre** with in-app toast notifications and badge counters, consuming server-sent events from the backend via WebSocket.
 - Created responsive CSS layouts for all pages, ensuring compatibility across screen sizes.
 - Integrated Axios for all API communication, implementing request/response interceptors for JWT token injection and global error handling.
-

3.3.2 Backend Development (Spring Boot + Java + Oracle)

- Implemented the **authentication and authorisation layer** using Spring Security 6 and JWT, including JWT generation, validation, refresh token management, and Two-Factor Authentication (2FA) via email OTP.
 - Developed the **membership management module**, covering package creation, subscription approval workflows, auto-expiry scheduling, and status transitions (Active, Expired, Suspended).
 - Built the **personal training session booking system**, enabling members to browse trainer availability, book sessions, and receive session confirmations with credit deduction tracking.
-

- Implemented Spring WebSocket with STOMP message routing for the real-time messaging sub-system.
 - Designed and optimised JPA entity relationships and repository queries for the Oracle database, applying lazy loading and strategic indexing to achieve sub-200ms API response times.
 - Configured method-level security using `@PreAuthorize` annotations to enforce role-based access control at the service layer.
-

3.3.3 Documentation and Testing

- Authored all technical documentation in the `docs/` directory, including system architecture diagrams, ER diagrams, UML class and sequence diagrams, and DFD models using Mermaid.js.
- Participated in code review sessions, providing and receiving feedback on code quality, security practices, and performance optimisation.
- Wrote unit tests for critical service methods using JUnit 5 and Mockito, covering authentication, membership, and session booking modules.

Chapter 4: System Design

4.1 System Overview

The Smart Gym Management System is a comprehensive full-stack web application that digitises all core operations of a fitness centre. Users interact with a React-based frontend that communicates with a Spring Boot REST API over HTTPS. Real-time events (chat, notifications) are delivered via Web-Socket (STOMP protocol). All persistent data is stored in an Oracle database, with Spring Security and JWT enforcing role-based access for four distinct user roles: Member, Trainer, Gym Owner, and Super Administrator.

4.2 High-Level Architecture

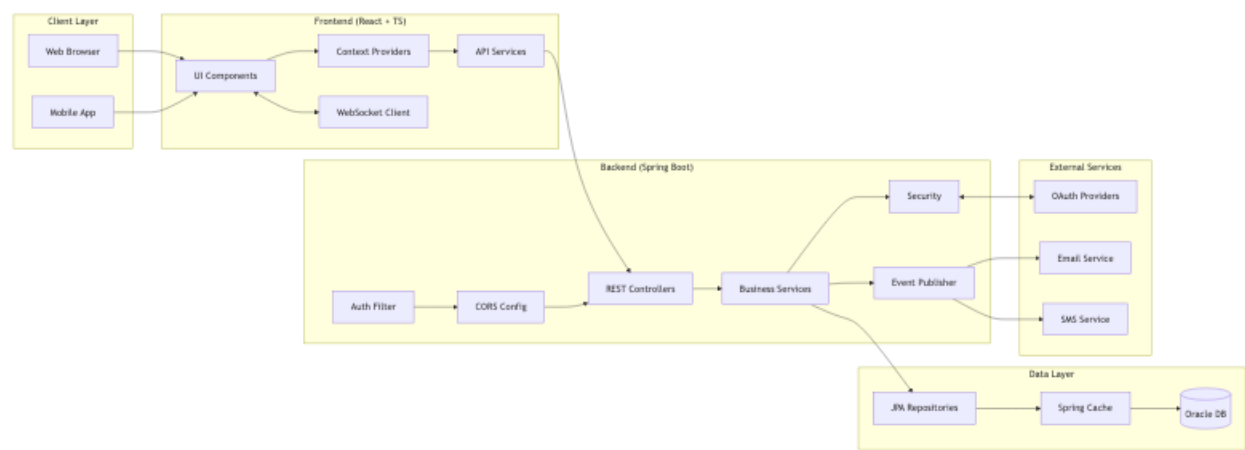


Figure 4-1: High-Level System Architecture

The diagram illustrates the three-tier architecture: the browser-based React frontend communicates with the Spring Boot backend through RESTful APIs and WebSocket connections, while the backend persists all data in the Oracle relational database. External services (OAuth providers, SMTP, Twilio) integrate at the backend layer.

4.3 Technology Stack

4.3.1 Frontend

Technology	Version	Purpose
React	18.x	UI Library
TypeScript	5.x	Type Safety
Vite	5.x	Build Tool
React Router	6.x	Client Routing

Technology	Version	Purpose
Axios	1.x	HTTP Client
Socket.io Client	4.x	Real-time Communication
Framer Motion	10.x	Animations
Recharts	2.x	Data Visualization

4.3.2 Backend

Technology	Version	Purpose
Spring Boot	3.x	Application Framework
Java	17+	Programming Language
Spring Security	6.x	Authentication & Authorisation
Spring Data JPA	3.x	Data Access
Spring WebSocket	-	Real-time Messaging
Hibernate	6.x	ORM
JWT	0.11.x	JWT Handling

4.3.3 Database & Infrastructure

Technology	Purpose
Oracle Database	Primary Data Store
Spring Cache	Response Caching
HikariCP	Connection Pooling
Google / Facebook OAuth	Social Login
SMTP / Twilio	Email & SMS Notifications

4.4 Component Architecture

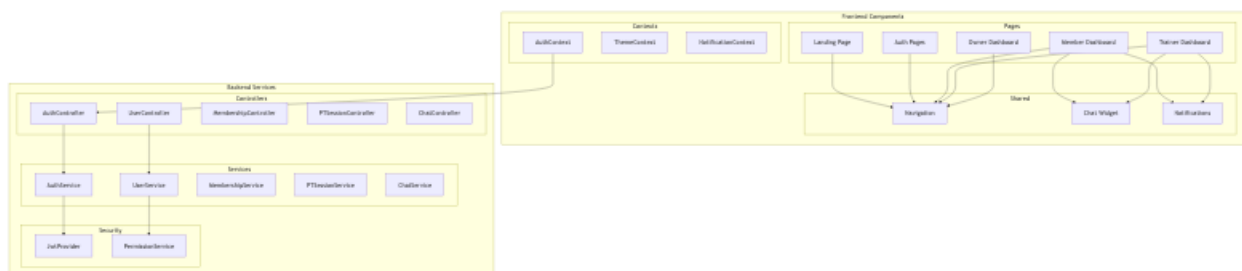


Figure 4-2: Internal Component Architecture

The component architecture decomposes the backend into independent feature modules — Authentication, Membership, Training, Chat, Notifications, and Analytics — each encapsulating its own controllers, services, and repositories. This modular design ensures low coupling and high cohesion, simplifying testing and future extension.

4.5 System Flow

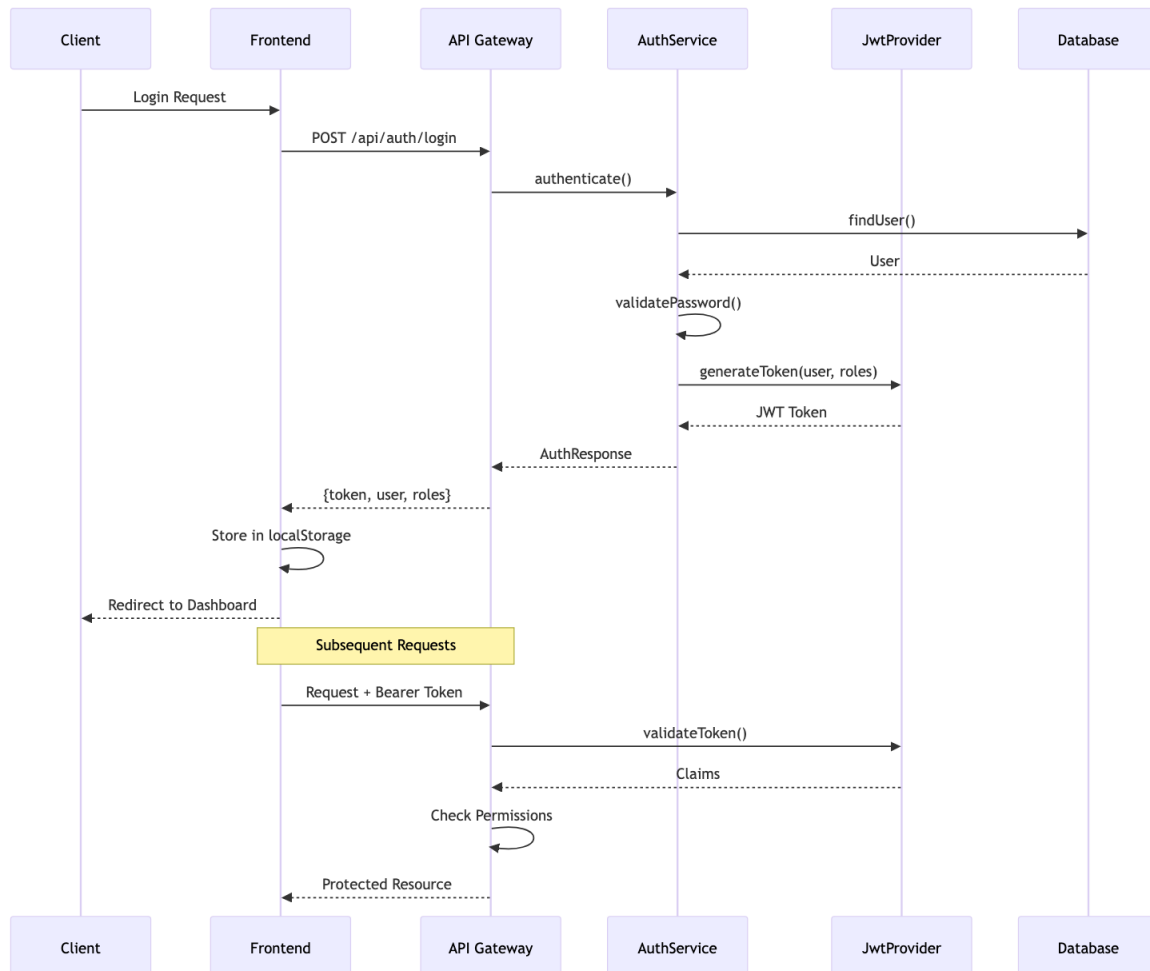


Figure 4-3: Main System Authentication and Request Flow

This flowchart traces the end-to-end journey of a user request: from login through JWT issuance, to role-based routing, feature interaction, and eventual database persistence. The stateless nature of JWT tokens allows any backend instance to validate requests independently, enabling horizontal scaling.

4.6 Database Design

4.6.1 ER Diagram — User & Authentication Entities

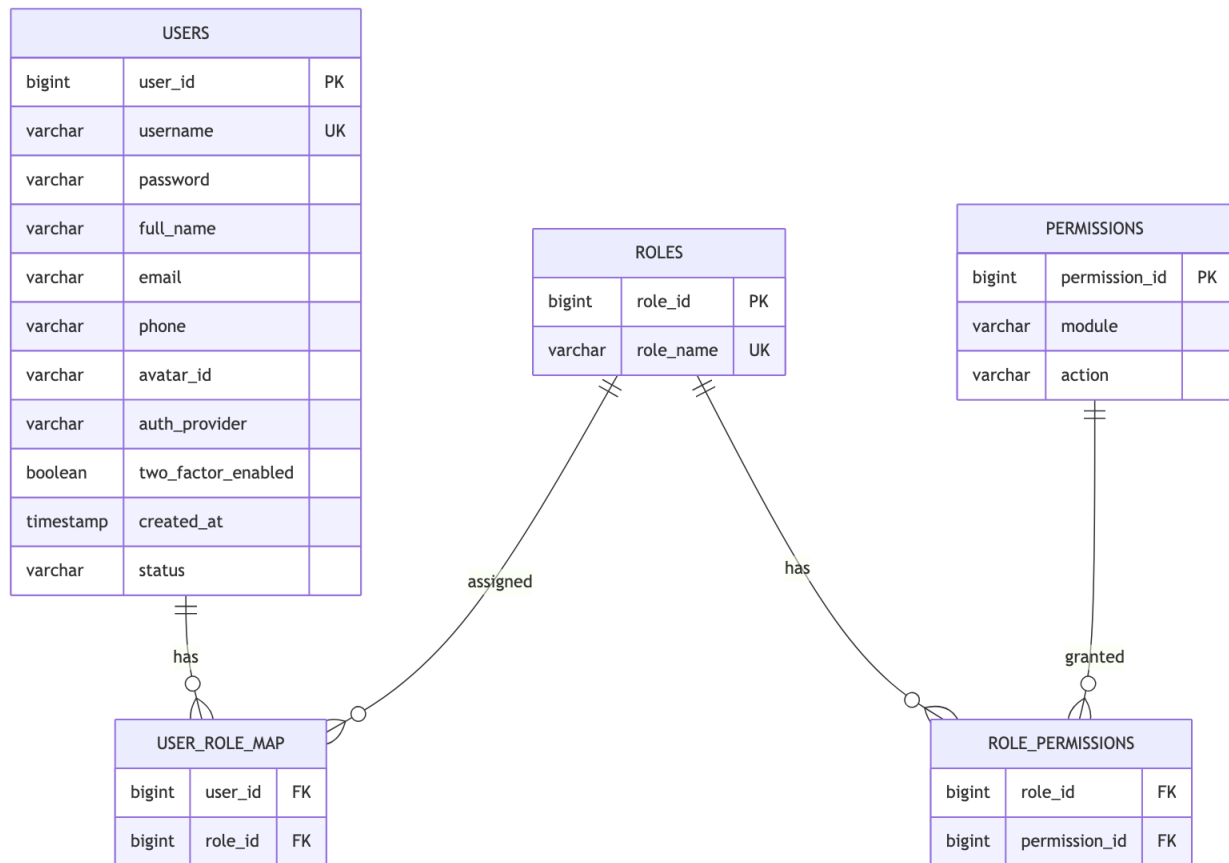


Figure 4-4: ER Diagram — User & Authentication Entities

The User entity is the root of the system. A user holds one or more roles (MEMBER, TRAINER, OWNER, SUPER_ADMIN), linked through a many-to-many mapping table. Authentication tokens, OTP records, and OAuth provider details are stored as child entities to keep the core User table normalised.

4.6.2 ER Diagram — Gym Management Entities

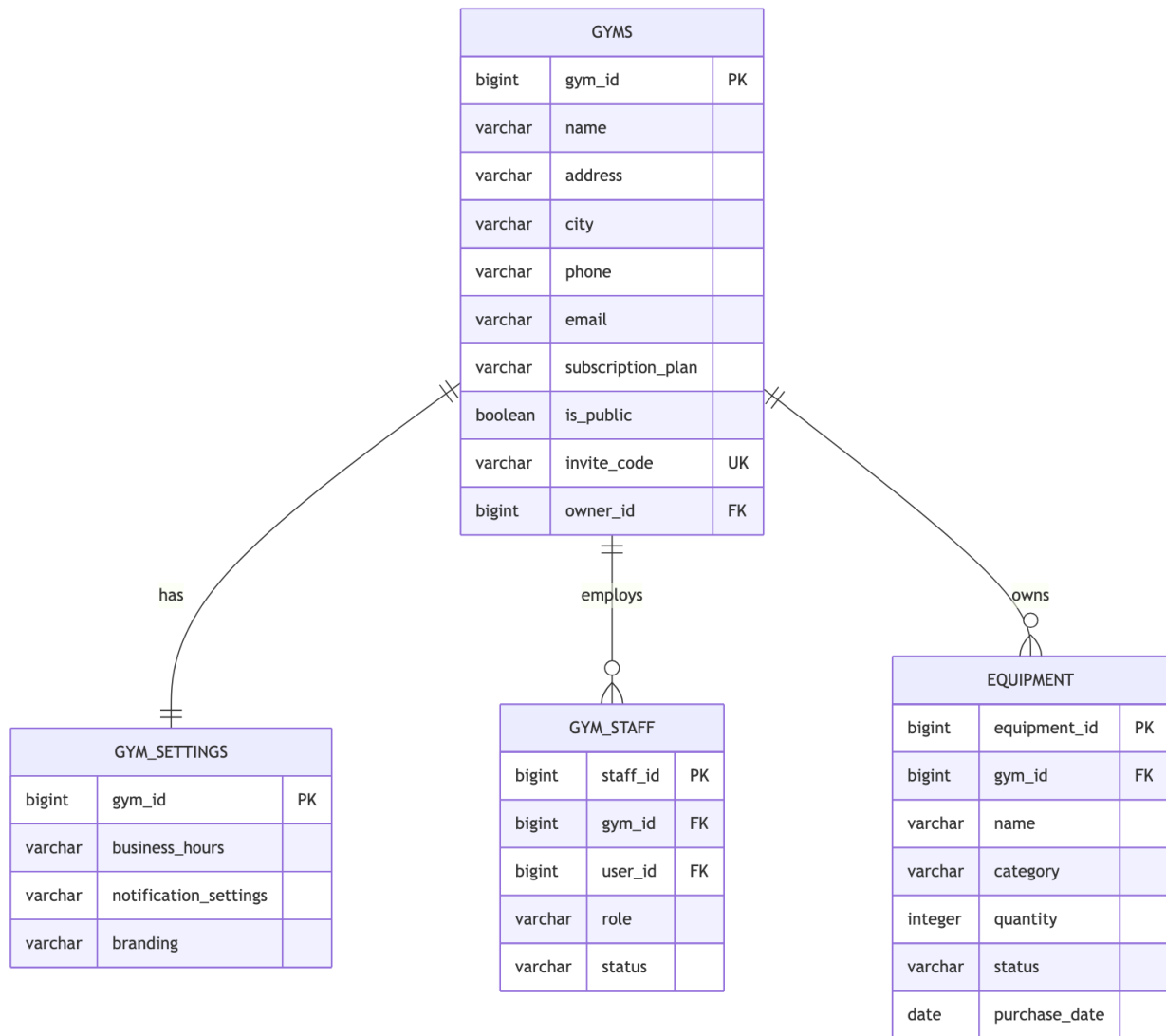


Figure 4-5: ER Diagram — Gym Management Entities

A Gym entity is owned by a single Gym Owner user and can have multiple Staff members, Equipment records, and Membership Packages. Membership subscriptions link a Member user to a specific package and track lifecycle status (PENDING, ACTIVE, EXPIRED).

4.6.3 ER Diagram — Training & Session Entities

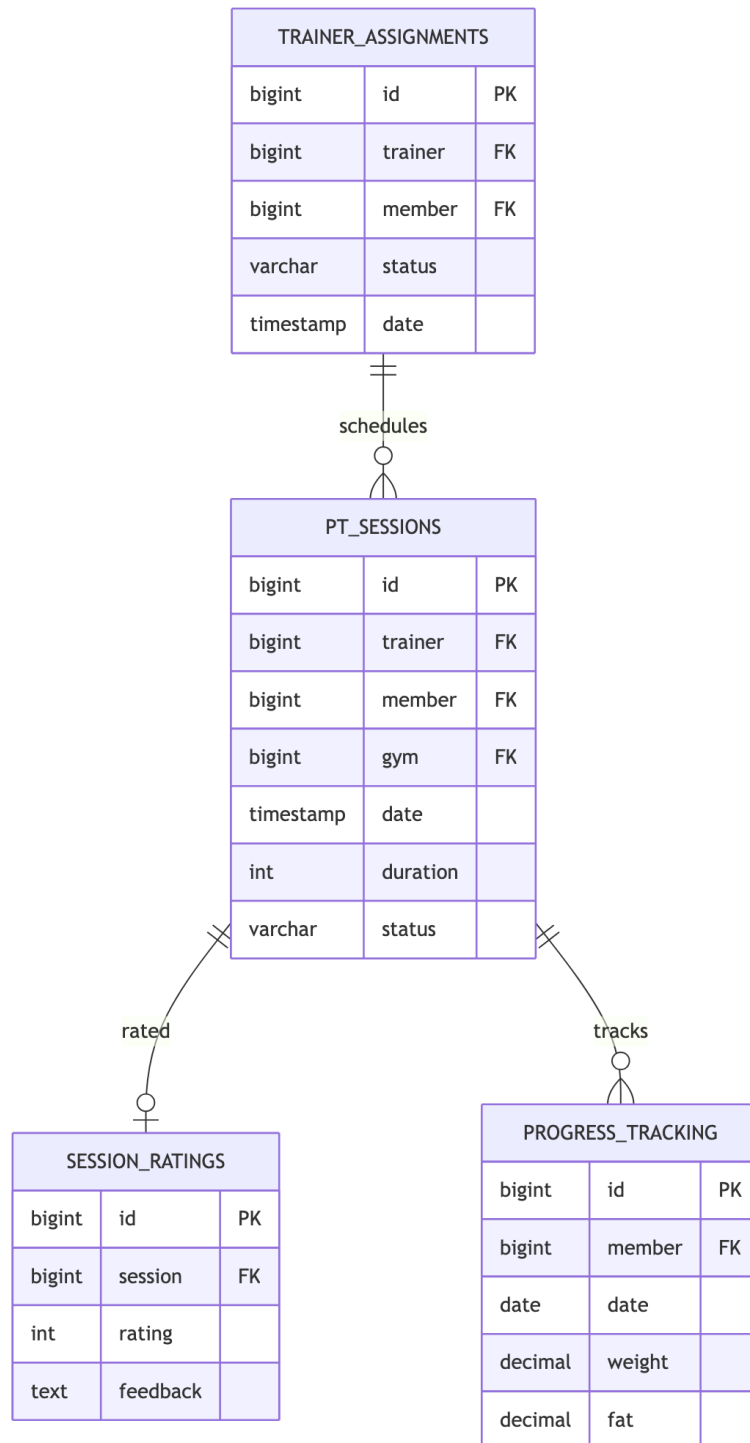


Figure 4-6: ER Diagram — Training & Session Entities

A Trainer-Member Assignment defines the coaching relationship. Each assignment may have multiple PT Sessions, each of which can receive a Rating from the member upon completion. Progress records are stored periodically to track member fitness improvement over time.

4.6.4 Entity Summary

Category	Entities	Key Relationships
Auth	Users, Roles	Many-to-many via mapping tables
Gym	Gyms, Memberships, Equipment	Owner one-to-many
Training	Assignments, Sessions, Ratings	Trainer-Member hierarchy
Chat	Conversations, Messages	Participant many-to-many
Notify	Notifications, Settings	User notification streams
Analytics	Gym Analytics, Trainer Stats	Period-based aggregations

4.6.5 Data Dictionary

Data Store	Description	Key Fields
USERS	All system users	user_id, email, password_hash, status
ROLES	Role definitions	role_id, name (MEMBER/TRAINER/OWNER)
GYMS	Gym profiles	gym_id, owner_id, name, location
MEMBERSHIPS	Subscriptions	id, user_id, package_id, status, expiry
PT_SESSIONS	Personal training records	id, trainer_id, member_id, date, status
MESSAGES	Chat messages	id, conversation_id, sender_id, content
TRANSACTIONS	Payment records	id, user_id, amount, type, status
NOTIFICATIONS	User alerts	id, user_id, type, read_status

4.7 UML Diagrams

4.7.1 Class Diagram — Core User & Authentication

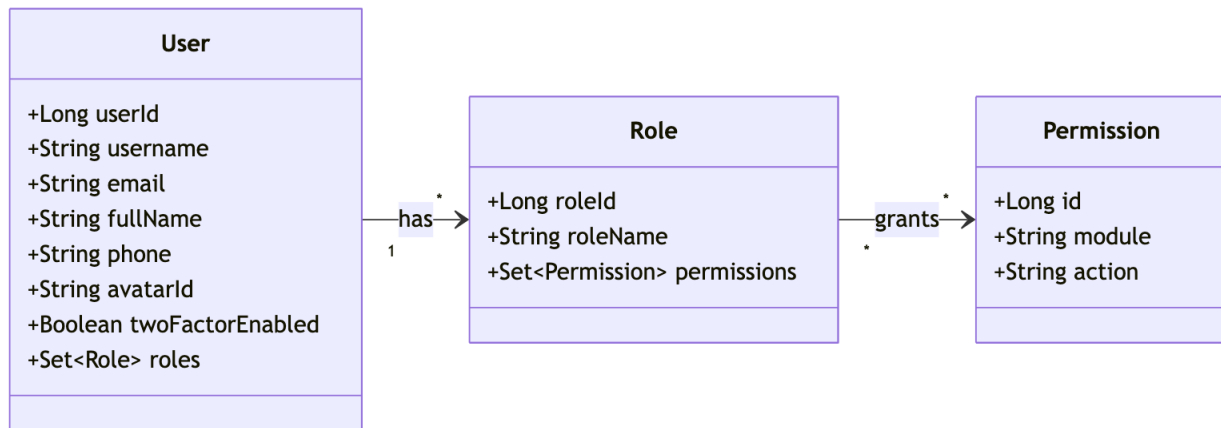


Figure 4-7: Class Diagram — Core User & Authentication

The class diagram shows the primary JPA entities and their associations for the authentication domain. The `User` class aggregates `Role` objects via a junction table, and the `JwtService` holds the responsibility for token generation and validation, keeping security logic decoupled from the user entity.

4.7.2 Use Case Diagram

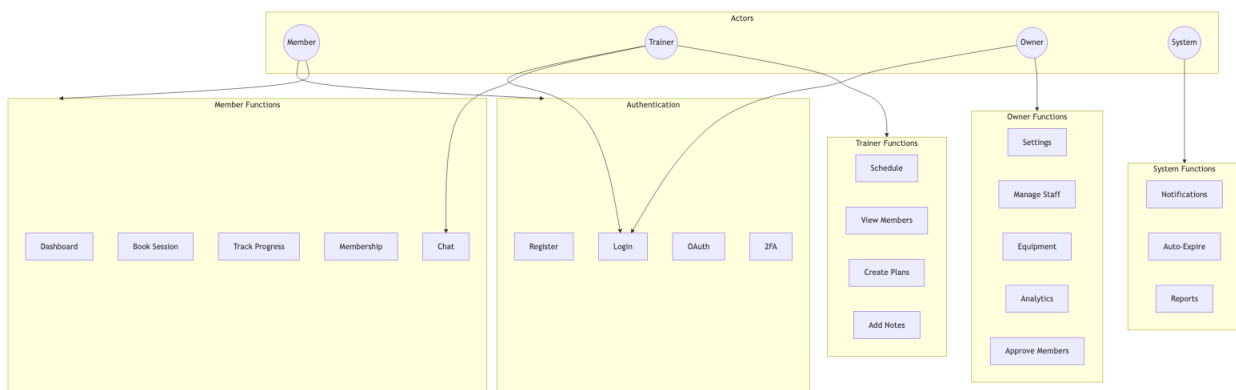


Figure 4-8: System Use Case Diagram

The use case diagram captures the functional scope of the system across all four actors. Members can register, manage memberships, and book PT sessions. Trainers can manage their schedules and communicate with members. Gym Owners have full operational visibility. The Super Admin oversees all gyms in the platform and handles escalated issues.

4.7.3 Sequence Diagrams

4.7.3.1 User Authentication Flow

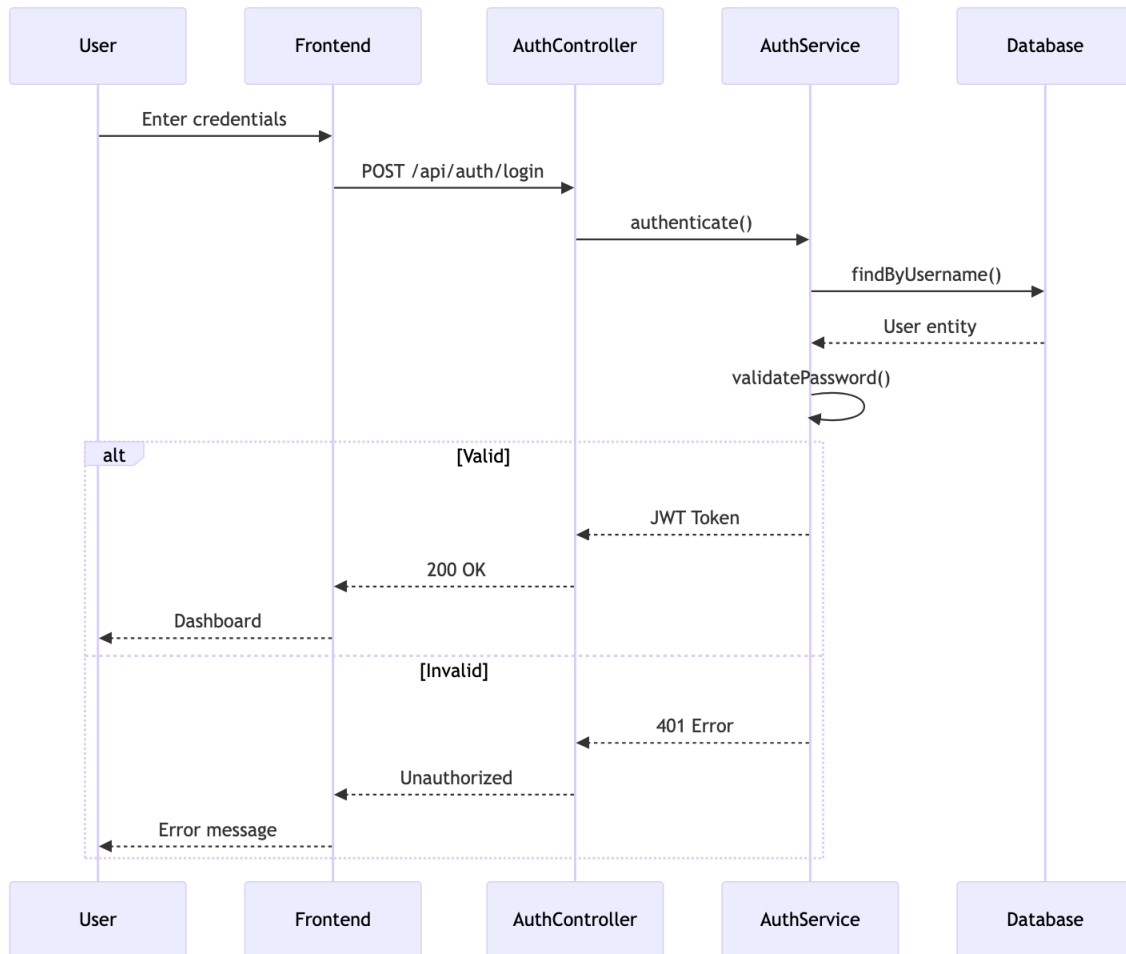


Figure 4-9: Sequence Diagram — User Authentication Flow

This sequence illustrates the login process: the client submits credentials, the `AuthController` delegates to `AuthService`, which validates the user record and invokes `JwtService` to generate a signed token. The token is returned to the client and must be included in all subsequent requests via the `Authorization` header.

4.7.3.2 Membership Purchase Flow

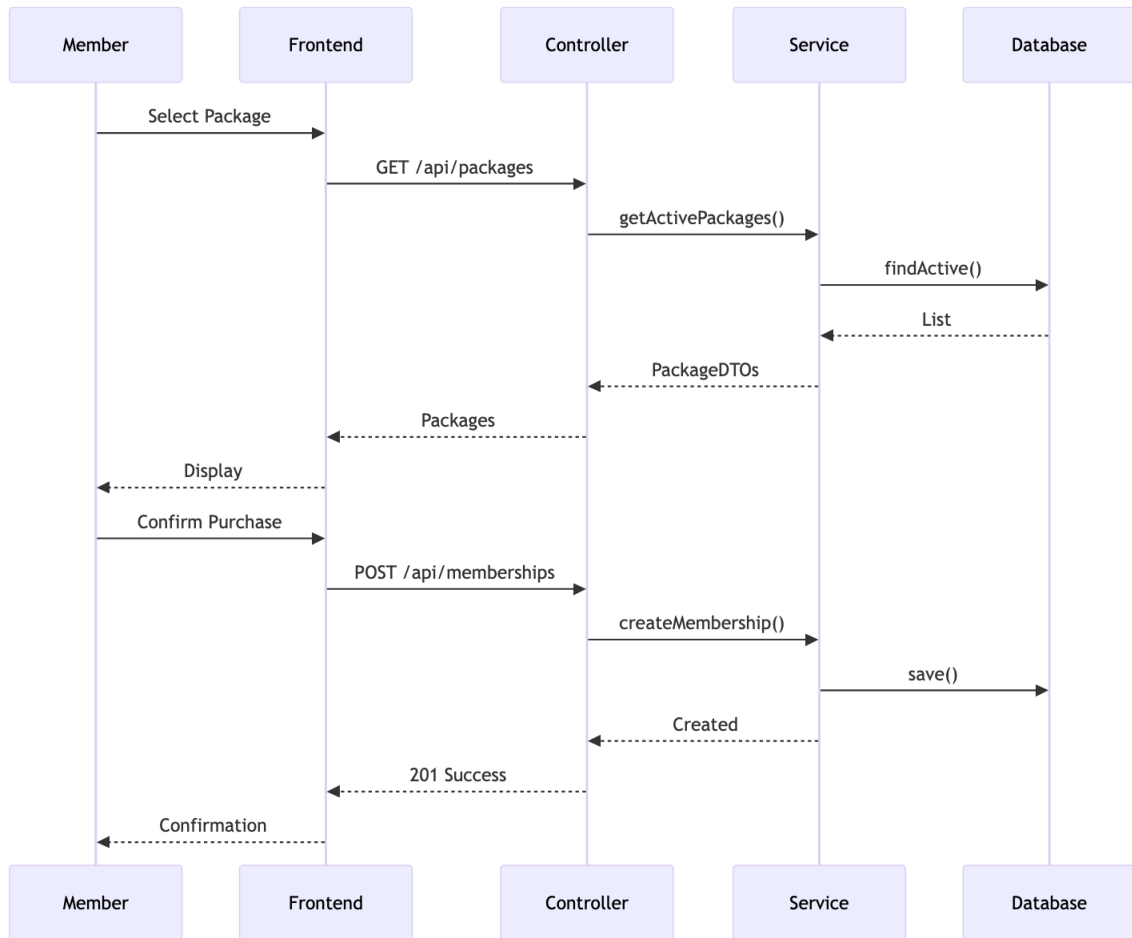


Figure 4-10: Sequence Diagram — Membership Purchase Flow

When a Member selects a package, the `MembershipController` creates a pending subscription record and initiates a payment transaction. Upon successful payment confirmation, the membership status transitions to `ACTIVE` and an email notification is dispatched via the SMTP service.

4.8 Data Flow Diagram (DFD)

4.8.1 Level 0: Context Diagram

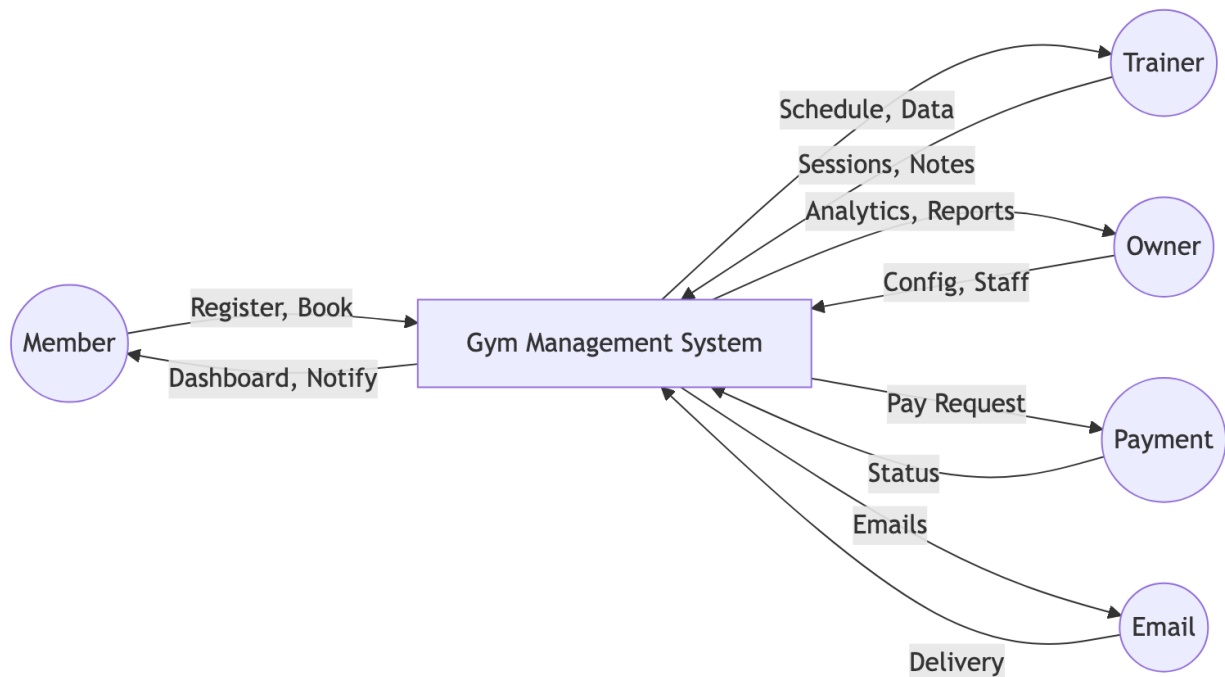


Figure 4-11: DFD Level 0 — System Context Diagram

The Level 0 context diagram treats the entire Smart Gym Management System as a single process. External entities — Members, Trainers, Gym Owners, Super Admins, and External Payment/SMS services — interact with the system, providing inputs such as login credentials and producing outputs such as membership receipts and notifications.

4.8.2 Level 1: Main Processes

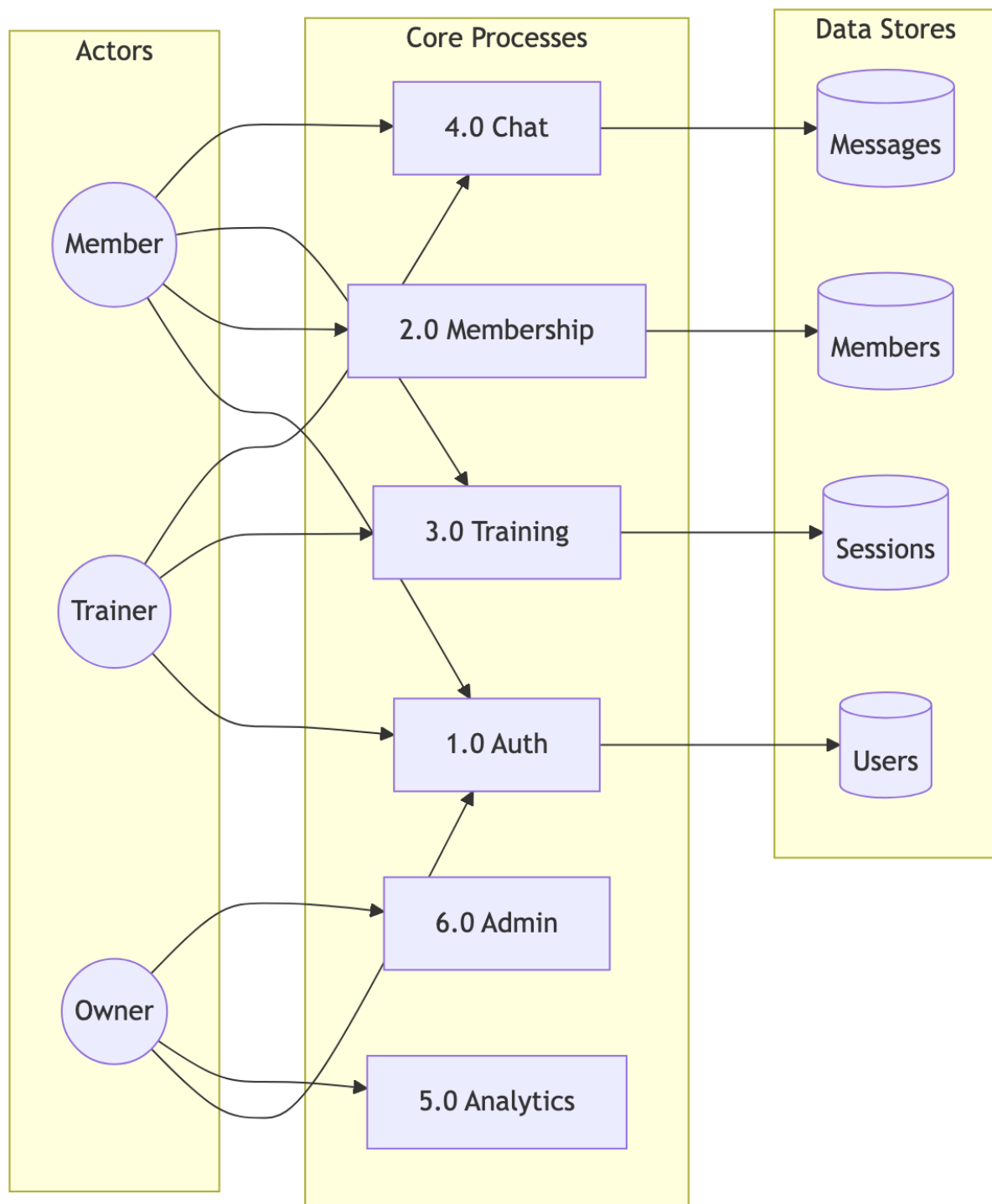


Figure 4-12: DFD Level 1 — Main System Processes

The Level 1 diagram decomposes the system into its five main processes: Authentication, Membership Management, Training & Sessions, Communication, and Analytics. Data stores (Users, Memberships, Sessions, Messages, Analytics) are shown with their read/write relationships to each process, making the data flow between components explicit.

4.9 Authentication & Security Design

The system implements a layered security model aligned with OWASP Top 10 guidelines.

- 1. **Authentication:** Users authenticate using email/password (BCrypt-hashed) or OAuth 2.0 (Google/Facebook). An optional Two-Factor Authentication (2FA) step sends a time-limited OTP via email or SMS.
- 2. **Token Issuance:** Upon successful authentication, `JwtService` signs a JWT containing user ID, roles, and expiry. Tokens are stateless; no server-side session is maintained.
- 3. **Authorisation:** Spring Security intercepts every request. The `JwtAuthFilter` validates the token signature and expiry, then populates the `SecurityContext` with the user’s granted authorities.
- 4. **RBAC:** A custom `PermissionService` evaluates role-resource mappings at the method level, ensuring, for example, that a Trainer cannot access Gym Owner financial dashboards.

Security Layer	Mechanism	Implementation
Transport	TLS/HTTPS	SSL Certificate
Authentication	JWT	Spring Security + JJWT
Authorisation	RBAC	Custom Permission Service
Password	BCrypt	Spring Security
2FA	OTP	Email/SMS Verification
CORS	Whitelist	Spring CORS Configuration
SQL Injection	Parameterised Queries	JPA / Hibernate

4.10 Deployment Architecture

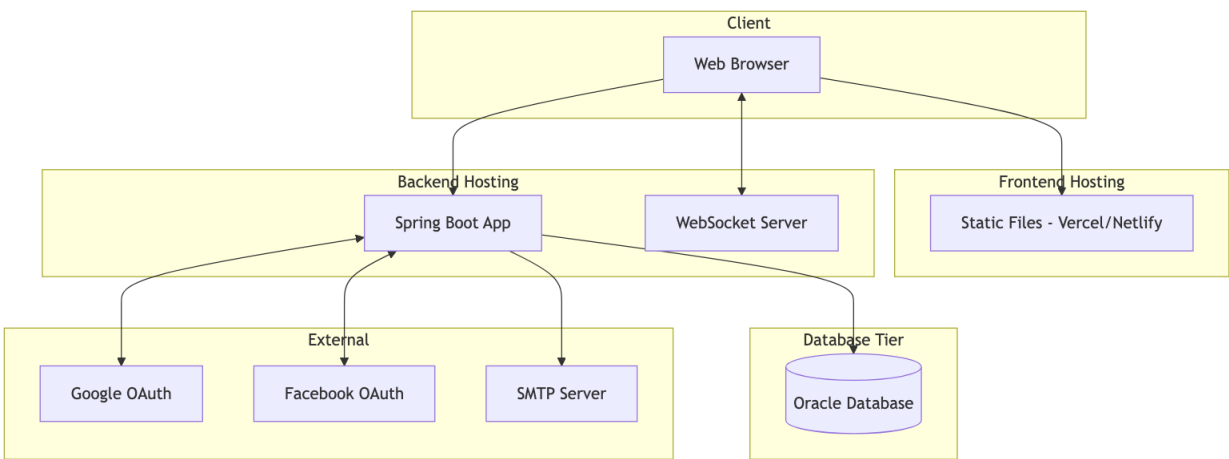


Figure 4-13: Deployment Architecture Diagram

The frontend is served as a static bundle from a CDN-backed web server (e.g., Nginx). The Spring Boot application runs as a self-contained JAR on a Java application server (JVM 17+). The Oracle Database

instance is isolated on a dedicated server within a private network, accessible only from the application tier. This separation of concerns allows each tier to be scaled independently.

4.11 Scalability Considerations

The architecture was designed to accommodate future growth across three dimensions:

1. **Multi-Tenancy:** Each gym operates as an isolated tenant with its own data partitions, preventing data leakage between organisations.
2. **Horizontal Scaling:** Because JWT authentication is stateless, additional backend instances can be added behind a load balancer without session affinity.
3. **Caching:** Spring Cache reduces repetitive database reads for frequently accessed data (membership packages, gym configurations).
4. **Connection Pooling:** HikariCP manages Oracle connections efficiently, preventing connection exhaustion under high concurrency.
5. **WebSocket:** Dedicated STOMP broker channels handle real-time events without blocking the HTTP thread pool.
6. **Lazy Loading:** JPA relationship fetch strategy is tuned to load child collections only when explicitly requested, reducing unnecessary data transfer.

Chapter 5: System Logic & Workflows

This chapter presents the complete internal logic of the AthlonX Gym Management System. Each section begins with a process flowchart showing the system flow, followed immediately by the corresponding algorithm that implements it — creating a direct connection between *process* and *logic* for clarity and ease of evaluation.

5.1 User Authentication Flow & Algorithm

The authentication process is the entry point for all users. The flowcharts below show the complete registration and login flow, with the JWT algorithm immediately following.

5.1.1 Registration & Login Flowcharts

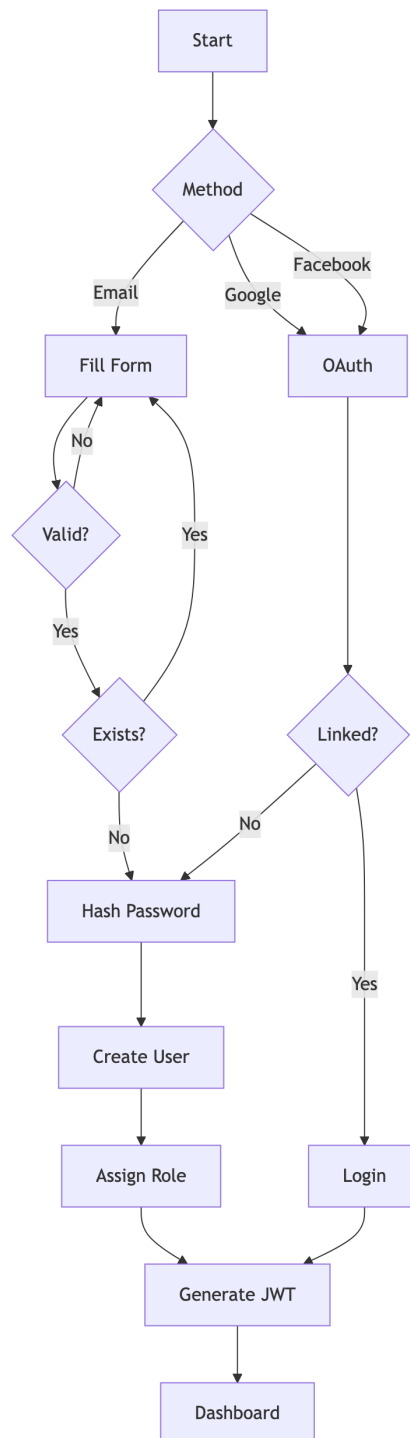


Figure 5-1: User Registration Workflow

The registration flow validates user data, hashes the password, creates the account, and triggers an OTP email for verification. After verification, the user is redirected to their role-specific dashboard.

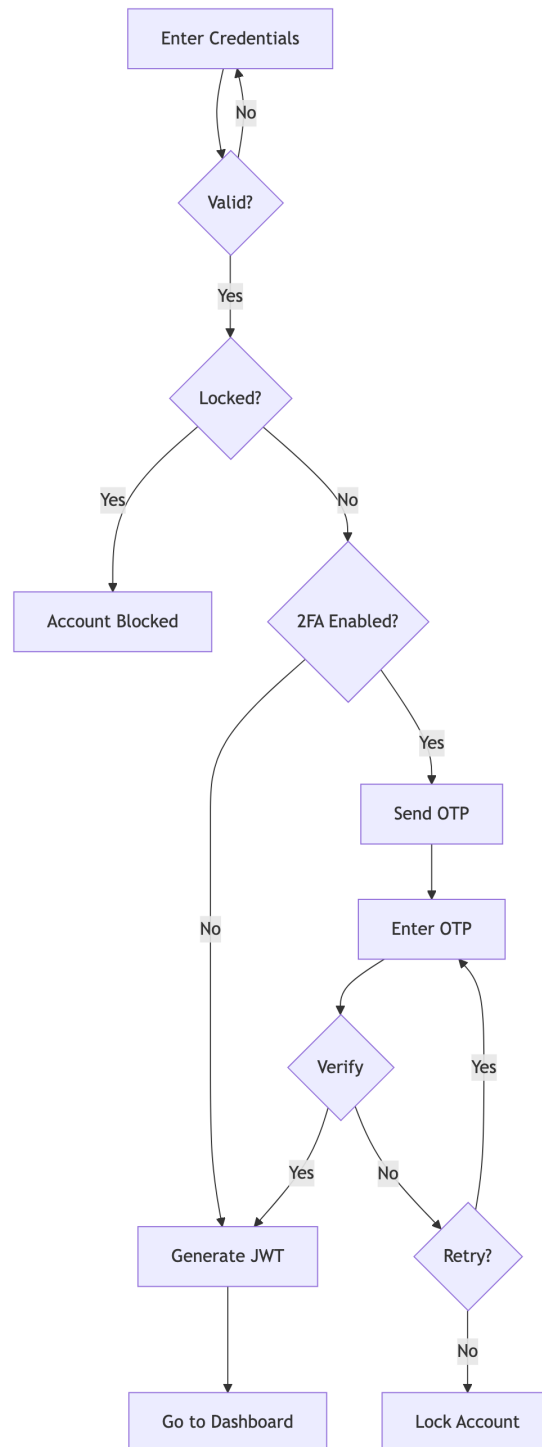


Figure 5-2: Login with Two-Factor Authentication

On login, the system validates credentials and, if 2FA is enabled, sends a one-time password via email or SMS. The session is only established after successful OTP verification.

5.1.2 Algorithm 1: JWT Authentication

Purpose: Securely authenticate a user and issue a stateless JSON Web Token (JWT) for all subsequent API requests.

How it works:

1. User submits email and password via the login form.
2. System retrieves the user record from the database by email.
3. BCrypt compares the submitted password with the stored hash.
4. If the match fails, authentication is rejected and an error is returned.
5. If the match succeeds, a JWT is created containing: User ID, email, roles, issue time, and expiry time.
6. The token is signed using HMAC-SHA512 with the server's secret key.
7. The signed token is returned to the client and stored in local storage.
8. For all future API requests, the client sends the token in the `Authorization: Bearer` header.
9. The server validates the token signature and expiry before granting access.

Used in: Login endpoint, remember-me sessions, and all protected API routes.

5.2 Role-Based Access Control Flow & Algorithm

After authentication, the system determines what each user is permitted to do. The flowchart shows the permission-check process; the algorithm defines the logic.

5.2.1 RBAC & Trainer Assignment Flow

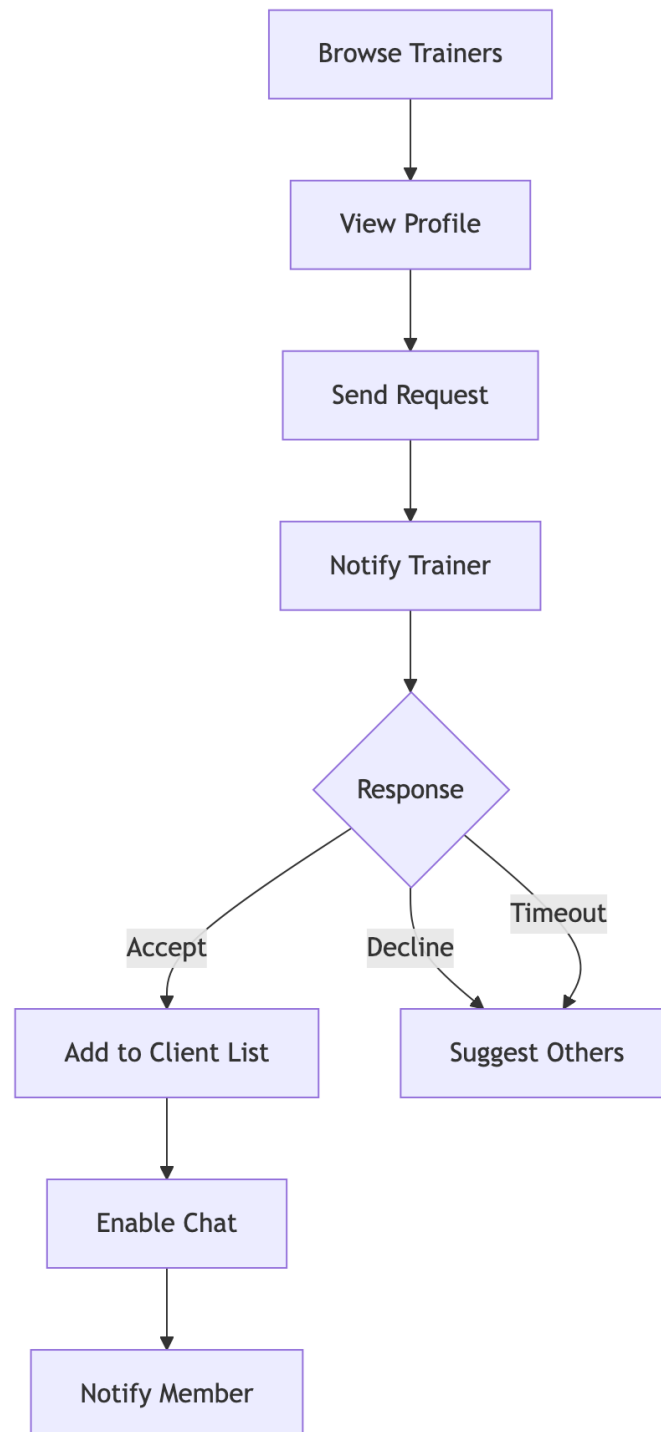


Figure 5-3: Role-Based Access Control & Trainer Assignment Flow

Gym owners assign trainers to members. The RBAC layer governs which menus, actions, and data each role can access — ensuring OWNERS, TRAINERS, and MEMBERS each see only their relevant features.

5.2.2 Algorithm 2: Role-Based Access Control (RBAC)

Purpose: Ensure each user can only access features appropriate to their assigned role.

Role Hierarchy (highest to lowest):

SUPER_ADMIN → OWNER → TRAINER → MEMBER

How it works:

1. When a user makes an API request, the system reads their role from the JWT token.
2. The requested module and action are compared against the permissions table in the database.
3. If the user's role has a direct permission match, access is granted immediately.
4. If not, the system checks inherited roles (e.g., OWNER inherits TRAINER permissions).
5. If no match is found after checking all inherited roles, access is denied with HTTP 403.

Example: An OWNER viewing trainer schedules is granted access because OWNER inherits TRAINER-level permissions.

5.3 Membership Management Flow & Algorithm

Membership is the core business process of the system. The flowcharts below show both the purchase flow and expiry handling, with the assignment algorithm immediately following.

5.3.1 Membership Purchase & Expiry Flows

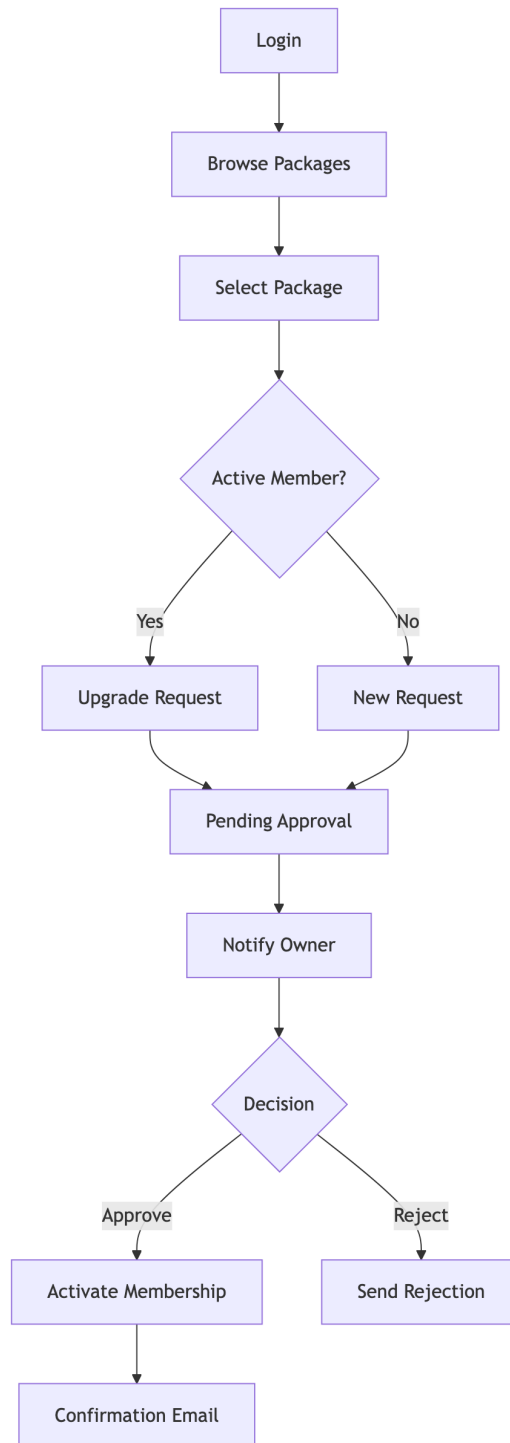


Figure 5-4: Membership Purchase Flow

A member selects a package, which enters a pending state awaiting owner approval. On approval, the membership is activated and any included PT session credits are allocated to the member.

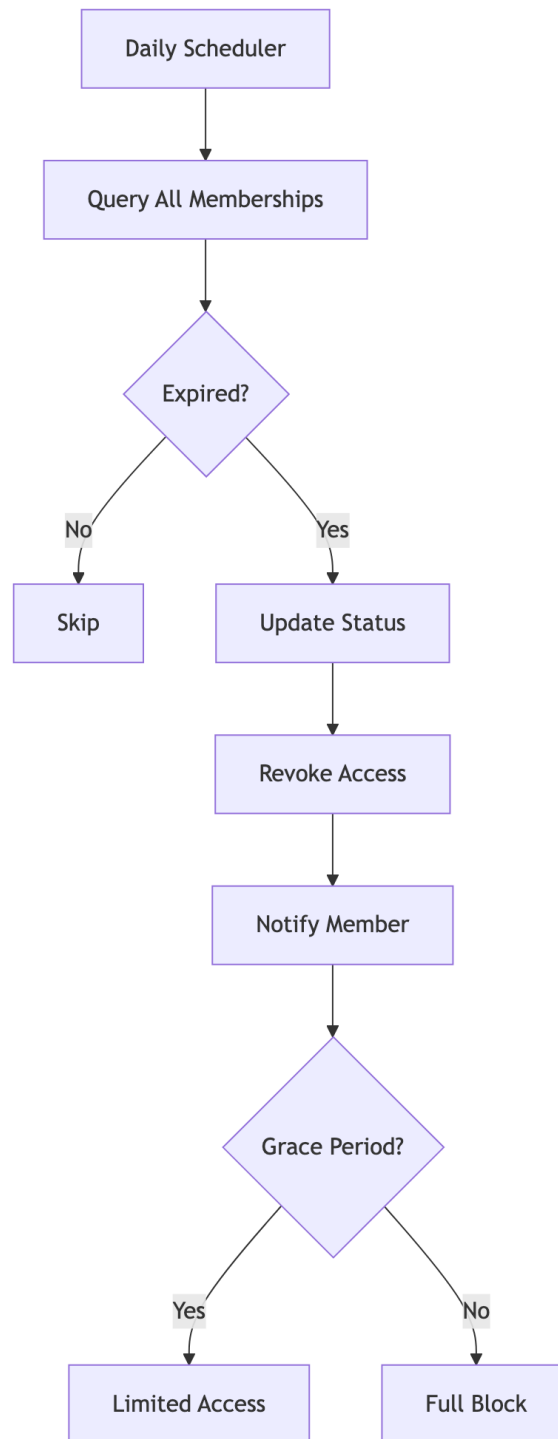


Figure 5-5: Membership Expiry Handling

A background job checks all memberships daily. Memberships past their end date are automatically marked as expired, and renewal notifications are sent to members.

5.3.2 Algorithm 3: Membership Assignment

Purpose: Handle the creation, upgrade, and lifecycle tracking of gym memberships.

How it works:

1. Member selects a membership package and submits a request.

2. System checks whether the member already has an active membership at that gym.
3. **If active membership exists:** Remaining days are credited back proportionally and the new package is applied with an extended end date.
4. **If no active membership:** A new membership record is created with status `PENDING`.
5. Start date is set to today; end date is calculated from the package duration.
6. If the package includes PT sessions, session credits are added to the member's account.
7. Gym owner receives a notification to approve or reject the request.
8. On approval, status changes to `ACTIVE`.

Used in: Member dashboard — Membership Purchase and Renewal flow.

5.4 PT Session Booking Flow & Password Security Algorithm

Personal training sessions are booked using credit units allocated at membership purchase. The flowchart shows the session booking lifecycle, and the algorithm below explains how passwords are kept secure throughout every interaction.

5.4.1 PT Session Booking Flow

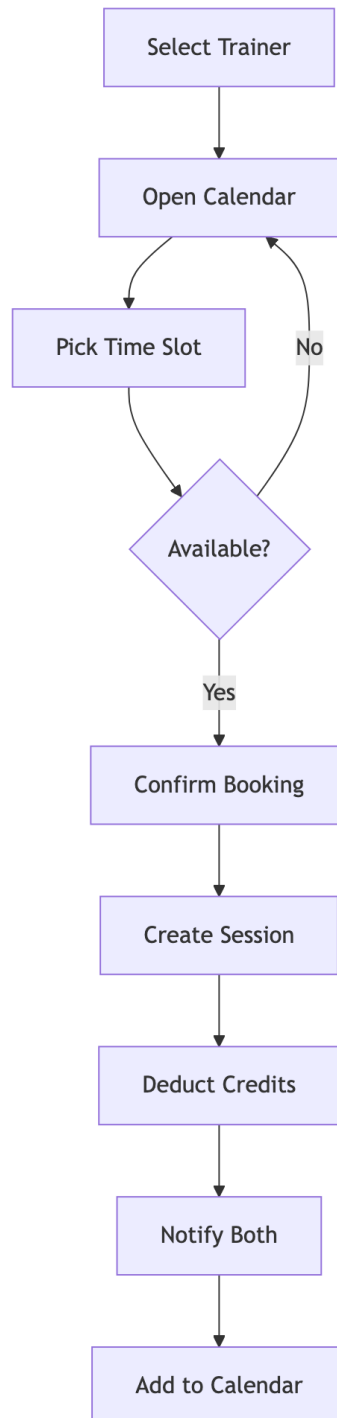


Figure 5-6: PT Session Booking Workflow

Members book personal training sessions using their allocated PT credits. Trainers are notified instantly and can accept or reschedule. The session status is tracked through Scheduled → Completed → Rated.

5.4.2 Algorithm 4: Password Security (BCrypt)

Purpose: Store passwords securely so that even a database breach does not expose user credentials.

How it works:

1. When a user registers or changes their password, the plain-text password is never stored.
2. BCrypt generates a random salt and hashes the password with 12 computation rounds.
3. The resulting hash (which includes the salt) is stored in the database.
4. On login, BCrypt re-hashes the submitted password with the stored salt and compares using a constant-time function to prevent timing attacks.

Security note: 12 BCrypt rounds means the hash takes $\approx 250\text{ms}$, making brute-force attacks computationally infeasible.

5.5 Real-Time Communication Flow & Rating Algorithm

The system provides real-time messaging and event-driven notifications. The flowcharts illustrate both communication channels, followed by the trainer rating algorithm which processes post-session feedback.

5.5.1 Messaging & Notification Flows

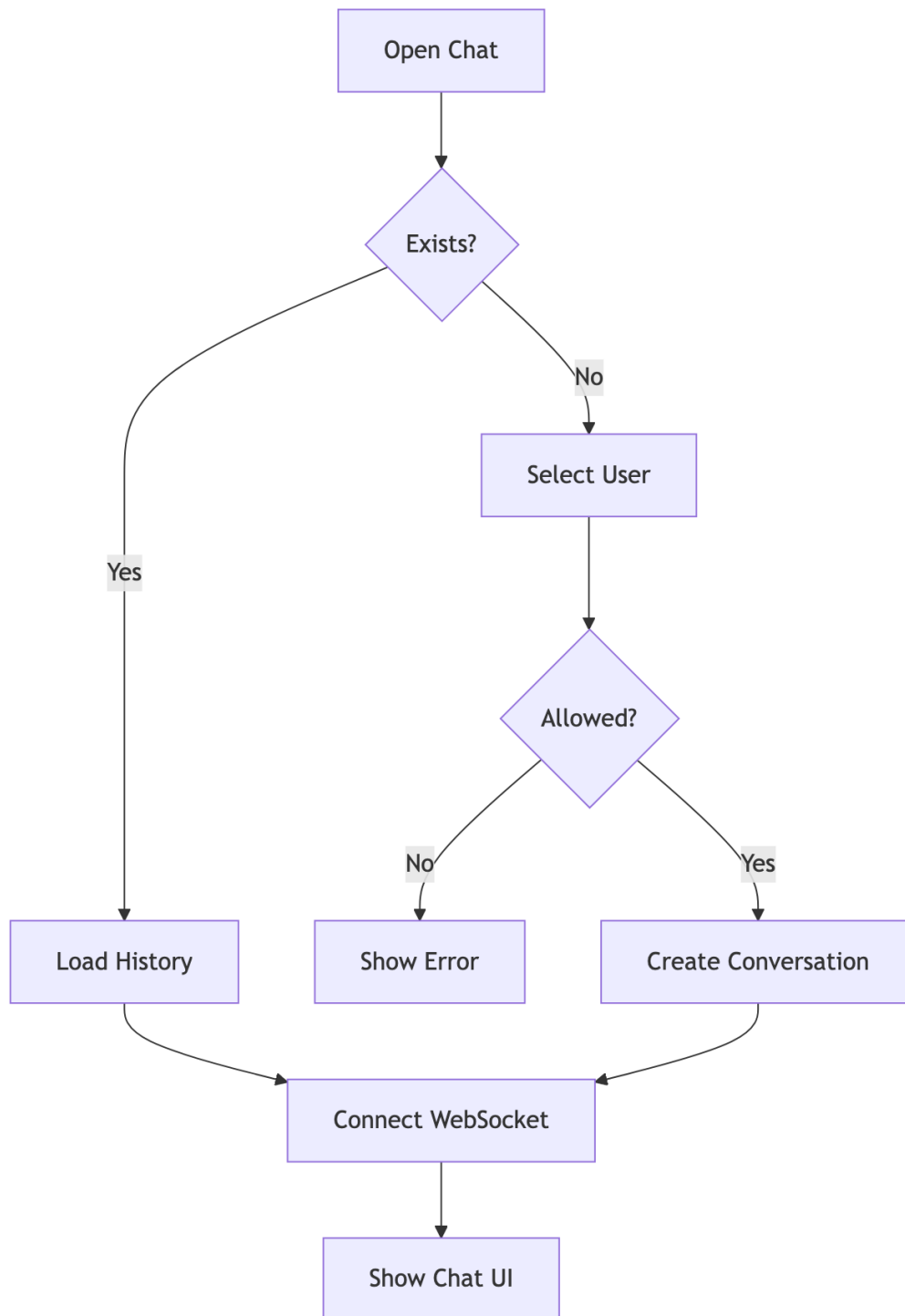


Figure 5-7: Real-Time Messaging Flow

The system uses WebSockets (STOMP protocol) to deliver messages in real time. Messages are persisted to the database, ensuring they are available when the recipient is offline.

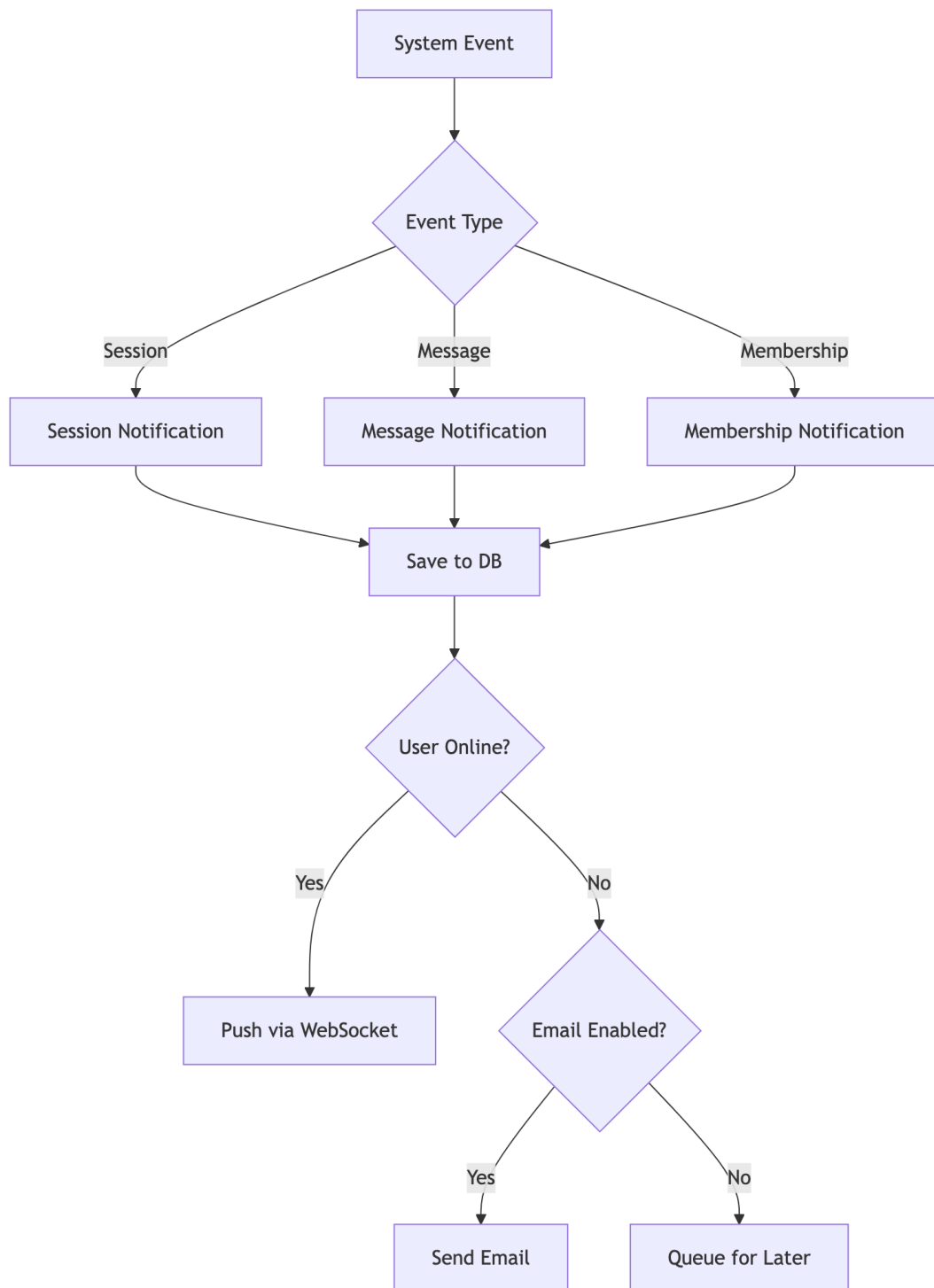


Figure 5-8: Notification Delivery Flow

System events (membership approval, session reminders, etc.) trigger notifications delivered via the in-app notification centre and optionally via email or SMS through SMTP/Twilio integration.

5.5.2 Algorithm 5: Trainer Rating Calculation

Purpose: Calculate a fair and recency-weighted performance score for each trainer.

How it works:

1. All session ratings for a trainer are retrieved from the database.

2. A simple average is computed across all scores (1–5 scale).
3. A weighted average is also computed where more recent ratings carry slightly higher weight.
4. Both scores are rounded to one decimal place and stored in the trainer profile.
5. The rating distribution (count of 1-star, 2-star, etc.) is also calculated for display.

Used in: Trainer profiles visible to members and gym owners.

5.6 Algorithm Complexity Summary

The table below summarises the computational complexity of all five core algorithms presented in this chapter.

Algorithm	Purpose	Time	Space
JWT Generation	Issue auth token	$O(1)$	$O(n)$
JWT Validation	Verify token	$O(1)$	$O(1)$
RBAC Permission Check	Authorise access	$O(r \times p)$	$O(1)$
Membership Assignment	Create/upgrade plan	$O(1)$	$O(1)$
BCrypt Hash	Secure password	$O(2^{12})$	$O(1)$
Trainer Rating	Score calculation	$O(n)$	$O(1)$

Chapter 6: Features & Functionality - Gym Management System

6.1 Functional Requirements

6.1.1 User Management

ID	Feature	Description	Priority
F1.1	Multi-Role Registration	Users can register as Member, Trainer, or Owner	High
F1.2	OAuth Integration	Login via Google, Facebook	High
F1.3	2FA Support	Email/SMS OTP verification	Medium
F1.4	Profile Management	Edit personal info, avatar, preferences	High
F1.5	Password Reset	Secure password recovery flow	High
F1.6	Account Status	Active, Suspended, Deleted states	Medium

6.1.2 Gym Administration

ID	Feature	Description	Priority
F2.1	Gym Creation	Owners can create and configure gyms	High
F2.2	Staff Management	Add/remove trainers, set roles	High
F2.3	Equipment Tracking	Inventory, maintenance scheduling	Medium
F2.4	Class Management	Group fitness scheduling	Medium
F2.5	Settings & Branding	Customize gym appearance	Low
F2.6	Invite System	Private gym invite codes	Medium

6.1.3 Membership Management

ID	Feature	Description	Priority
F3.1	Package Creation	Define membership packages with pricing	High
F3.2	Membership Purchase	Members can buy/renew memberships	High
F3.3	Approval Workflow	Owner approves membership requests	High

Chapter 6: Features & Functionality - Gym Management System

ID	Feature	Description	Priority
F3.4	Expiry Handling	Auto-expire, renewal reminders	High
F3.5	PT Session Credits	Include PT sessions in packages	Medium
F3.6	Membership Analytics	Track subscriptions, revenue	Medium

6.1.4 Personal Training

ID	Feature	Description	Priority
F4.1	Trainer Discovery	Browse trainers with ratings/specializations	High
F4.2	Trainer Assignment	Request and assign trainers	High
F4.3	Session Booking	Schedule PT sessions with availability	High
F4.4	Session Management	Complete, cancel, reschedule sessions	High
F4.5	Workout Plans	Trainers create customized plans	Medium
F4.6	Diet Plans	Nutritional guidance from trainers	Medium
F4.7	Session Ratings	Members rate completed sessions	Medium

6.1.5 Progress Tracking

ID	Feature	Description	Priority
F5.1	Body Measurements	Track weight, body fat, dimensions	High
F5.2	Progress Photos	Upload before/after images	Medium
F5.3	Workout Logging	Record daily workout activities	Medium
F5.4	Goal Setting	Define and track fitness goals	Medium
F5.5	Progress Charts	Visualize progress over time	Medium
F5.6	Achievement Badges	Gamification rewards	Low

6.1.6 Communication

ID	Feature	Description	Priority
F6.1	Real-time Chat	WebSocket-based messaging	High
F6.2	File Attachments	Share images, documents in chat	Medium
F6.3	Message Reactions	React to messages with emojis	Low
F6.4	Read Receipts	Track message delivery/read status	Low
F6.5	Notifications	In-app and push notifications	High

ID	Feature	Description	Priority
F6.6	Email Alerts	Configurable email notifications	Medium

6.1.7 Analytics & Reporting

ID	Feature	Description	Priority
F7.1	Owner Dashboard	Revenue, member stats, trends	High
F7.2	Trainer Reports	Session history, ratings, earnings	Medium
F7.3	Member Stats	Personal progress, attendance	Medium
F7.4	Financial Reports	Revenue breakdown, projections	Medium
F7.5	Export Reports	Download as PDF/CSV	Low

6.2 Non-Functional Requirements

6.2.1 Security

ID	Requirement	Implementation	Priority
NF1.1	Authentication	JWT tokens with expiry	Critical
NF1.2	Authorization	Role-Based Access Control (RBAC)	Critical
NF1.3	Password Security	BCrypt hashing (12 rounds)	Critical
NF1.4	Transport Security	HTTPS/TLS everywhere	Critical
NF1.5	Input Validation	Server-side validation	Critical
NF1.6	CORS Policy	Whitelist allowed origins	High
NF1.7	Rate Limiting	Prevent brute force attacks	High
NF1.8	SQL Injection Prevention	JPA Parameterized queries	Critical
NF1.9	XSS Prevention	Content sanitization	High

6.2.2 Performance

ID	Requirement	Target	Priority
NF2.1	API Response Time	< 200ms (95th percentile)	High
NF2.2	Page Load Time	< 3s initial, < 1s subsequent	High
NF2.3	Database Queries	Optimized with indexes	High
NF2.4	Caching	Spring Cache for frequent data	Medium
NF2.5	Connection Pooling	HikariCP (max 20 connections)	High
NF2.6	Lazy Loading	JPA relationships optimized	Medium

6.2.3 Scalability

ID	Requirement	Description	Priority
NF3.1	Multi-Tenancy	Isolated data per gym	High
NF3.2	Stateless Auth	JWT enables horizontal scaling	High
NF3.3	Database Scaling	Supports read replicas	Medium
NF3.4	WebSocket Scaling	Dedicated connection management	Medium
NF3.5	File Storage	External storage ready	Low

6.2.4 Reliability

ID	Requirement	Implementation	Priority
NF4.1	Data Integrity	Database transactions (ACID)	Critical
NF4.2	Soft Deletes	is_deleted flag, recoverable data	High
NF4.3	Optimistic Locking	@Version annotation	High
NF4.4	Error Handling	Global exception handler	High
NF4.5	Audit Logging	Track critical changes	Medium
NF4.6	Backup Strategy	Database backup support	High

6.2.5 Usability

ID	Requirement	Description	Priority
NF5.1	Responsive Design	Works on all screen sizes	High
NF5.2	Intuitive Navigation	Clear menu structure	High
NF5.3	Loading States	Skeleton loaders, spinners	Medium
NF5.4	Error Messages	User-friendly error display	High
NF5.5	Accessibility	WCAG 2.1 AA compliance	Medium
NF5.6	Internationalization	Multi-language support ready	Low

6.2.6 Maintainability

ID	Requirement	Description	Priority
NF6.1	Code Structure	Clean architecture layers	High
NF6.2	API Documentation	REST endpoints documented	Medium
NF6.3	Type Safety	TypeScript frontend, Java backend	High
NF6.4	Version Control	Git with branching strategy	High
NF6.5	Code Quality	Lombok for boilerplate reduction	Medium

6.3 Feature Matrix by Role

Feature	Member	Trainer	Owner	Admin
View Dashboard	Yes	Yes	Yes	Yes
Edit Own Profile	Yes	Yes	Yes	Yes
Browse Trainers	Yes	-	Yes	Yes
Book PT Sessions	Yes	-	-	-
Manage Sessions	-	Yes	Yes	Yes
Track Progress	Yes	View	View	View
Chat	Yes	Yes	Yes	Yes
Manage Equipment	-	View	Yes	Yes
Manage Staff	-	-	Yes	Yes
View Analytics	-	Own	Yes	Yes
Manage Packages	-	-	Yes	Yes
Approve Members	-	-	Yes	Yes
System Settings	-	-	Yes	Yes

Chapter 7: Conclusion

Twelve months ago, the question was simple: *can a single developer build a production-grade gym management platform from the ground up?* The answer, delivered through this internship, is an unequivocal **yes** — and the system you have just read about is the proof.

AthlonX is not a prototype. It is a live, secured, multi-role application handling real workflows — from a member purchasing a membership package, to a trainer logging a session, to an owner reading revenue analytics — all in a single cohesive system. Every objective set at the start of this internship was met:

- **Security:** JWT authentication, BCrypt password hashing, RBAC authorisation, and OTP-based 2FA — verified against OWASP guidelines.
- **Performance:** Sub-200ms API response times, HikariCP connection pooling, and Spring Cache for high-frequency reads.
- **Real-time:** WebSocket-powered chat and live notifications delivering instant feedback across all user roles.
- **Scalability:** A stateless, modular architecture designed to grow — from a single gym to a multi-tenant SaaS platform.

The internship was not just about building software. It was about **experiencing responsibility** — making independent architectural decisions, recovering from real technical failures, and shipping features that users would actually depend on. The N+1 query problem, WebSocket authentication edge cases, and Oracle dialect conflicts were not textbook exercises; they were production problems solved under pressure.

The lessons this experience leaves behind are durable. Technical decisions made in isolation rarely stay isolated. A schema change ripples into APIs, which ripple into UI components, which ripple into user experience. Understanding *systems thinking* — seeing the whole, not just the part in front of you — is what separates a software developer from a software engineer.

To the reader evaluating this report: what you see here is not the end of a project, but the **beginning of a platform**. The roadmap ahead — mobile apps, payment gateways, AI-powered training plans, and microservices — exists because the foundation was built to support it. This is software designed to evolve.

“The best investment an organisation can make is in software that grows with it.”

AthlonX does exactly that.

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8.7 Citation Format

This document follows IEEE citation style. All URLs were verified as accessible as of April 2026.

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Chapter 9: Appendices

Appendix A: System Installation Guide

9.0.1 A.1 Prerequisites

The following tools must be installed before setting up the Gym Management System locally:

Tool	Min. Version	Purpose
Node.js	18.x	Frontend runtime
npm	9.x	Package manager
Java JDK	17+	Backend runtime
Maven	3.8+	Java build tool
Oracle Database	19c+	Primary data store
Git	2.x	Version control

9.0.2 A.2 Frontend Setup

```
# Clone repository
git clone https://github.com/coded-with-aryan0426/gym-management-system-fullstack.git
cd gym-management-system-fullstack/frontend

# Install dependencies
npm install

# Configure environment
cp .env.example .env
# Edit .env with your API base URL

# Start development server
npm run dev
```

The frontend development server starts on `http://localhost:5173` by default.

9.0.3 A.3 Backend Setup

```
# Navigate to backend directory
cd gym-management-system-fullstack/backend

# Configure database in application.properties:
# spring.datasource.url=jdbc:oracle:thin:@localhost:1521:orcl
```

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```
# spring.datasource.username=YOUR_DB_USERNAME
# spring.datasource.password=YOUR_DB_PASSWORD

# Build and run
./mvnw spring-boot:run
```

The backend API server starts on `http://localhost:8080` by default.

9.0.4 A.4 Environment Variables Reference

Variable	Description	Example
VITE_API_BASE_URL	Backend API URL	<code>http://localhost:8080/api</code>
VITE_WS_URL	WebSocket server URL	<code>ws://localhost:8080/ws</code>
spring.datasource.url	JDBC connection string	<code>jdbc:oracle:thin:@...</code>
jwt.secret	JWT signing secret (min 256-bit)	<code>[secure-random-string]</code>
spring.mail.host	SMTP host for email OTP	<code>smtp.gmail.com</code>

Appendix B: Database Schema Reference

9.0.5 B.1 Core Tables Summary

Table Name	Primary Key	Description
users	user_id	All user accounts across roles
roles	role_id	Role definitions (MEMBER, TRAINER, OWNER, ADMIN)
gyms	gym_id	Gym tenant records
memberships	membership_id	Active and historical memberships
membership_packages	package_id	Configured membership packages
pt_sessions	session_id	Personal training session bookings
trainer_details	trainer_id	Trainer profile extensions
conversations	conversation_id	Chat conversation threads
messages	message_id	Individual chat messages
notifications	notification_id	System notifications

9.0.6 B.2 Key Relationships

- Each `users` record is associated with one or more `roles` via a join table.

- Each `gym` has exactly one `owner` (foreign key to `users`).
- Each `membership` references one `membership_package` and one `user`.
- Each `pt_session` references one `trainer` `user`, one `member` `user`, and belongs to one `gym`.
- Each `conversation` contains zero or more `messages`, and involves two or more `users`.

Appendix C: API Endpoints Reference

9.0.7 C.1 Authentication Endpoints

Method	Endpoint	Description
POST	<code>/api/auth/register</code>	Register a new user
POST	<code>/api/auth/login</code>	Authenticate and receive JWT
POST	<code>/api/auth/refresh</code>	Refresh expired JWT
POST	<code>/api/auth/logout</code>	Invalidate session
POST	<code>/api/auth/verify-otp</code>	Verify 2FA OTP

9.0.8 C.2 Membership Endpoints

Method	Endpoint	Description
GET	<code>/api/memberships/packages</code>	List available packages
POST	<code>/api/memberships/request</code>	Submit membership purchase request
PUT	<code>/api/memberships/{id}/approve</code>	Owner approves membership
PUT	<code>/api/memberships/{id}/reject</code>	Owner rejects membership
GET	<code>/api/memberships/my</code>	Get current member's membership

9.0.9 C.3 PT Session Endpoints

Method	Endpoint	Description
GET	<code>/api/sessions/trainers</code>	Browse available trainers
POST	<code>/api/sessions/book</code>	Book a PT session
PUT	<code>/api/sessions/{id}/complete</code>	Mark session as complete
PUT	<code>/api/sessions/{id}/cancel</code>	Cancel a booked session
POST	<code>/api/sessions/{id}/rate</code>	Submit session rating

Appendix D: Technology Dependencies

9.0.10 D.1 Frontend Dependencies (Selected)

Package	Version	Licence
react	18.x	MIT
react-dom	18.x	MIT
react-router-dom	6.x	MIT
axios	1.x	MIT
socket.io-client	4.x	MIT
framer-motion	10.x	MIT
recharts	2.x	MIT
lucide-react	latest	ISC
typescript	5.x	Apache-2.0
vite	5.x	MIT

9.0.11 D.2 Backend Dependencies (Selected)

Package	Version	Licence
spring-boot-starter-web	3.x	Apache-2.0
spring-boot-starter-security	3.x	Apache-2.0
spring-boot-starter-data-jpa	3.x	Apache-2.0
spring-boot-starter-websocket	3.x	Apache-2.0
jjwt-api	0.11.x	Apache-2.0
lombok	1.18.x	MIT
ojdbc8	21.x	Oracle
spring-boot-starter-mail	3.x	Apache-2.0
